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ENGINEERING MANAGEMENT

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SUPPLY CHAIN MANAGEMENT & TRANSPORTATION MODEL

Introduction to Supply Chain

⇒ A **supply chain** is the sequence of organizations, their facilities, functions, and activities involved in producing and delivering a product or service.

- The sequence **starts from basic suppliers of raw materials**
- Extends **up to the final customers**

❖ Facilities in a Supply Chain

- Warehouses
- Factories
- Processing centers
- Distribution centers
- Retail outlets
- Offices

❖ Functions & Activities

- Forecasting
- Purchasing
- Inventory management
- Information management
- Quality assurance
- Scheduling
- Production
- Distribution
- Delivery
- Customer service

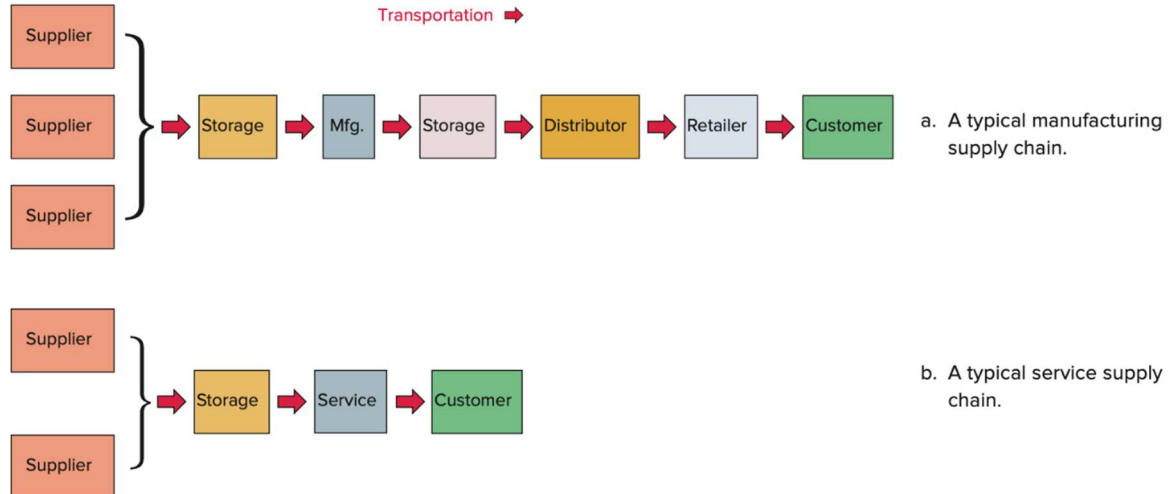


Fig: Typical Supply Chain

Flow Management in Supply Chain

⇒ An important aspect of supply chain management is **flow management**.

❖ Three Types of Flows

1. Product & Service Flow

- Movement of goods/services from suppliers to customers
- Handling customer service
- Product returns

2. Information Flow

- Sharing forecast and sales data
- Transmitting orders
- Tracking shipments
- Updating order status

3. Financial Flow

- Credit terms
- Payments
- Consignment
- Title/ownership arrangements

Logistics & Logistics Management

⇒ **Logistics** refers to the movement of:

- Materials
- Services
- Cash
- Information

within a supply chain.

❖ Materials Include

- Raw materials
- Work-in-process (WIP)
- Finished goods
- Support items such as:
 - Fuel
 - Equipment
 - Parts
 - Tools
 - Lubricants
 - Office supplies

❖ Logistics Includes

- Movement within a facility
- Incoming shipments
- Outgoing shipments
- Information flow throughout the supply chain

Movement Within a Facility

⇒ Movement of goods inside a manufacturing facility is part of **production control**.

❖ Steps of Movement

1. Incoming vehicles → Receiving
2. Receiving → Storage
3. Storage → Point of use (Work center)
4. Work center → Work center / Temporary storage
5. Final operation → Final storage
6. Storage → Packaging / Shipping
7. Shipping → Outgoing vehicles

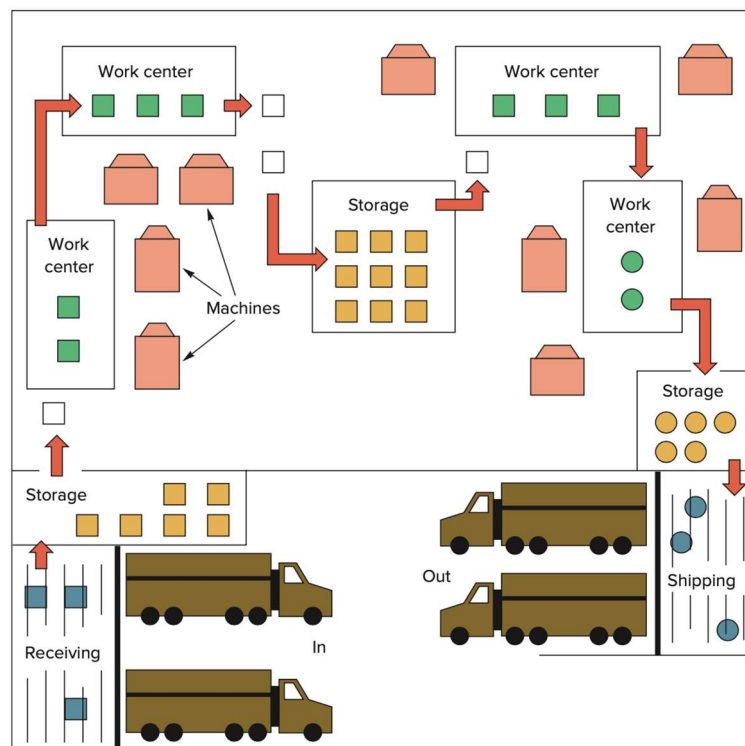
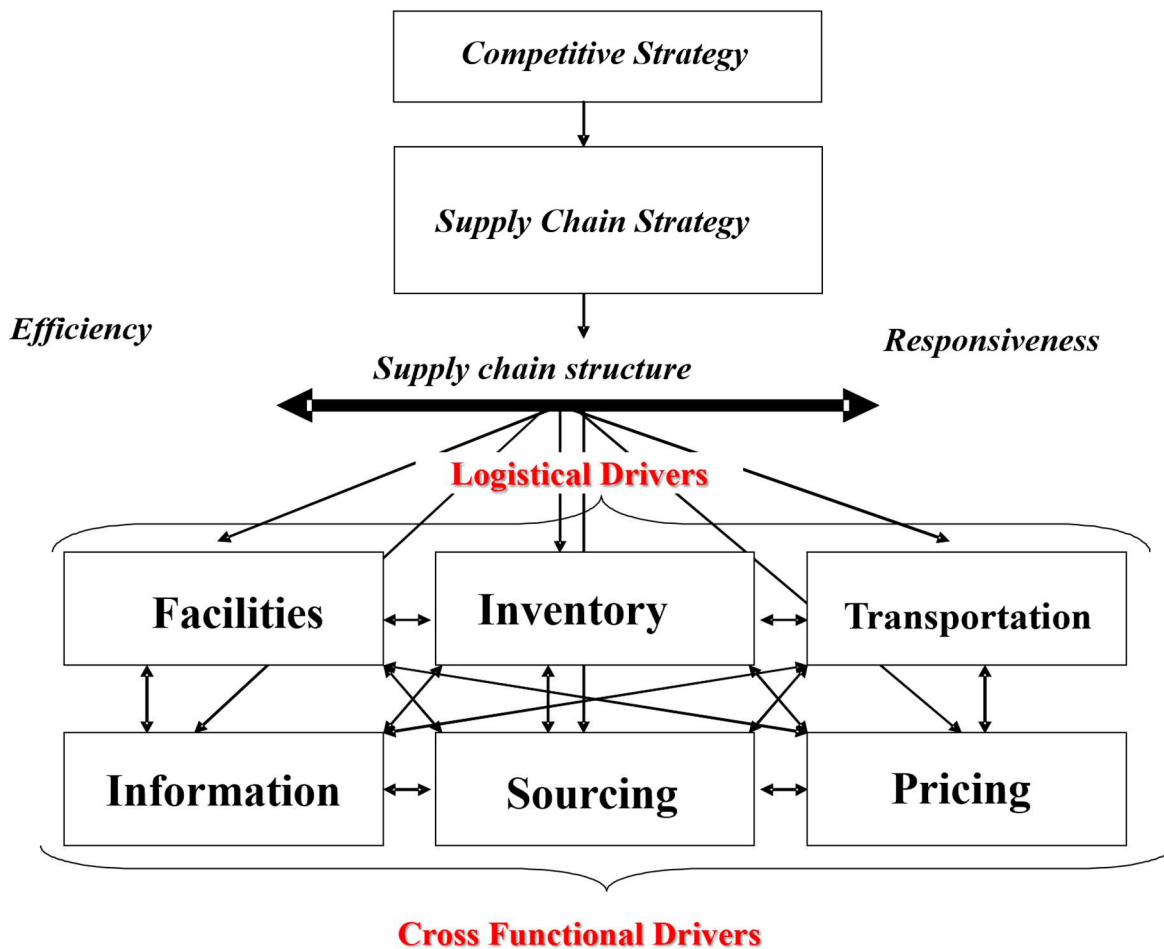


Fig: Movement Within a Facility

Difference Between Logistic and Supply Chain Management



A Framework for Structuring Drivers



Drivers of Supply Chain Performance

1. Facilities

- Places where inventory is:
 - Stored
 - Assembled
 - Fabricated
- Includes production sites and storage sites

2. Inventory

- Raw materials
- Work-in-process (WIP)
- Finished goods
- Inventory policies

3. Transportation

- Moving inventory from point to point
- Selection of transportation modes and routes

4. Information

- Data regarding:
 - Inventory
 - Transportation
 - Facilities
- **Biggest driver** of supply chain performance

5. Sourcing

- Functions performed by a firm
- Functions outsourced to others

6. Pricing

- Price of goods and services provided to the supply chain

Transportation Model

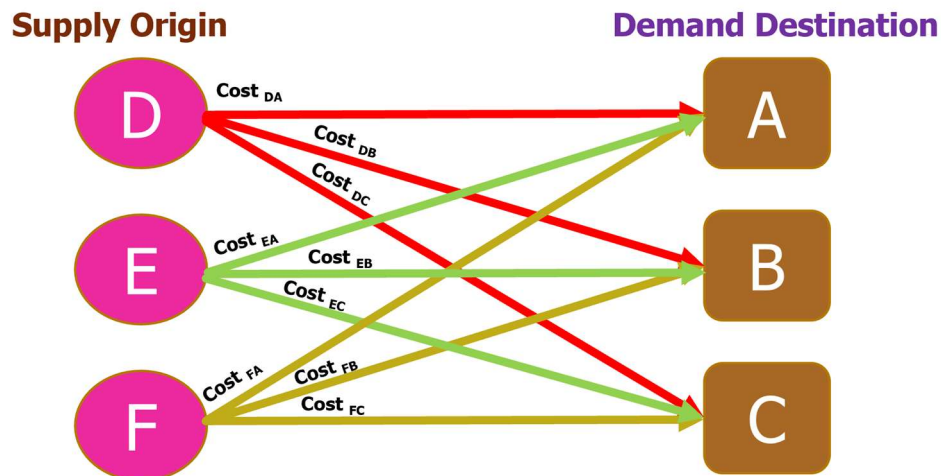
- ⇒ A **Transportation Model** is a special case of **Linear Programming Problem (LPP)**.
- ⇒ The objective **Transportation Model** To transport products from **multiple sources** to **multiple destinations** at **minimum total cost**.

❖ Aim of the Transportation Model:

- Find the **optimum transportation schedule**
- Minimize transportation cost

Structure of Transportation Model

- Supply origins: D, E, F
- Demand destinations: A, B, C
- Transportation cost exists between every source and destination



Types of Transportation Problems

1. Balanced Transportation Problem

- $\sum Supply = \sum Demand$

2. Unbalanced Transportation Problem

- $\sum Supply \neq \sum Demand$

Methods to Solve Transportation Model

1. North-West Corner Method (NWCM)
2. Least Cost Method (LCM)
3. Vogel's Approximation Method (VAM)

Note: Least Cost Method does **not always** provide minimum cost. Final solution requires optimization techniques.

Least Cost Method (LCM)

❖ Steps:

1. Check whether the problem is balanced
2. Choose the cell with **least transportation cost**
3. Allocate maximum possible units
4. Adjust supply and demand
5. Repeat until allocation is complete
6. Calculate total transportation cost

✂ **Problem 1:** A mobile phone manufacturing company has three production branches located at **Barishal, Rajshahi, and Rangpur**. The company supplies mobile phones to three cities: **Dhaka, Sylhet, and Chittagong**.

The available supply from Barishal, Rajshahi, and Rangpur is **40 units, 60 units, and 70 units** respectively. The demand at Dhaka, Sylhet, and Chittagong is **70 units, 40 units, and 60 units** respectively.

The **transportation cost per unit (in Tk.)** from each source to each destination is given in the following cost matrix:

Sources \ Destinations	Dhaka	Sylhet	Chittagong
Barishal	4	5	1
Rajshahi	3	4	6
Rangpur	6	2	8

- (a) Formulate the given problem as a transportation problem.
- (b) Check whether the transportation problem is balanced.
- (c) Using the **Least Cost Method**, determine a **Basic Feasible Solution (BFS)**.
- (d) Calculate the **total transportation cost**.

(a)

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1	40
Rajshahi	3	4	6	60
Rangpur	6	2	8	70
Demand	70	40	60	170/170

(b)

Since,

$$\sum Supply = \sum Demand = 170$$

So,

Balanced Transportation Problem

(c)

❖ Choose the cell with **least transportation cost**:

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1 ₍₄₀₎	40 ₍₀₎
Rajshahi	3	4	6	60
Rangpur	6	2	8	70
Demand	70	40	60 ₍₂₀₎	170/170

❖ Now again, Choose the cell with **least transportation cost**:

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1 ₍₄₀₎	40 ₍₀₎
Rajshahi	3	4	6	60
Rangpur	6	2 ₍₄₀₎	8	70 ₍₃₀₎
Demand	70	40 ₍₀₎	60 ₍₂₀₎	170/170

❖ Now again, Choose the cell with **least transportation cost**:

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1 ₍₄₀₎	40 ₍₀₎
Rajshahi	3 ₍₆₀₎	4	6	60 ₍₀₎
Rangpur	6	2 ₍₄₀₎	8	70 ₍₃₀₎
Demand	70 ₍₁₀₎	40 ₍₀₎	60 ₍₂₀₎	170/170

❖ Now again, Choose the cell with **least transportation cost**:

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1 ₍₄₀₎	40 ₍₀₎
Rajshahi	3 ₍₆₀₎	4	6	60 ₍₀₎
Rangpur	6 ₍₁₀₎	2 ₍₄₀₎	8	70 _{(30) (20)}
Demand	70 _{(10) (0)}	40 ₍₀₎	60 ₍₂₀₎	170/170

❖ Now again, Choose the cell with **least transportation cost**:

Sources \ Destinations	Dhaka	Sylhet	Chittagong	Supply
Barishal	4	5	1 ₍₄₀₎	40 ₍₀₎
Rajshahi	3 ₍₆₀₎	4	6	60 ₍₀₎
Rangpur	6 ₍₁₀₎	2 ₍₄₀₎	8 ₍₂₀₎	70 _{(30) (20) (0)}
Demand	70 _{(10) (0)}	40 ₍₀₎	60 _{(20) (0)}	170/170

(d)

⇒ **We Know**

$$\text{Total Cost} = \Sigma (\text{Allocated Units} \times \text{Cost per Unit})$$

⇒ **Total Transportation Cost:**

$$\begin{aligned}
 \text{Total Cost} &= (40 \times 1) + (60 \times 3) + (10 \times 6) + (40 \times 2) + (20 \times 8) \\
 &= 40 + 180 + 60 + 80 + 160 \\
 &= \text{Tk. } 520
 \end{aligned}$$

Unbalanced Transportation Problem

- If **Supply < Demand** → Add a **Dummy Row** to balance the transportation problem.
- If **Supply > Demand** → Add a **Dummy Column** to balance the transportation problem.
- Transportation cost for dummy cells = **0**

Note: After balancing, the problem can be solved using **NWCM, LCM, or VAM**, and the **total transportation cost** is calculated by:

$$\text{Total Cost} = \sum (\text{Allocated Units} \times \text{Cost per Unit})$$

✂ **Problem 2:** A manufacturing company has three factories located at **A, B, and C**, which supply goods to three warehouses located at **W₁, W₂, W₃**.

The supply available at factories **A, B, and C** is **20 units, 70 units, and 25 units** respectively. The demand at warehouses **W₁, W₂, W₃** is **65 units, 30 units, and 50 units** respectively.

The transportation cost per unit (in Tk.) from each factory to each warehouse is given in the following table:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1	7	20
B	7	4	6	70
C	3	2	5	25
Demand	65	30	50	

- Determine whether the transportation problem is balanced or unbalanced.
- Obtain a **Basic Feasible Solution (BFS)** using the **Least Cost Method**.
- Calculate the **total transportation cost**.

(a)

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1	7	20
B	7	4	6	70
C	3	2	5	25
Demand	65	30	50	145/115

Since,

$$\sum \text{Supply} \neq \sum \text{Demand}$$

So,

Unbalanced Transportation Problem

(b)

Since **Supply < Demand** → Add a **Dummy Row** to balance the transportation problem.

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1	7	20
B	7	4	6	70
C	3	2	5	25
Dummy	0	0	0	30
Demand	65	30	50	145/145

❖ Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7	4	6	70
C	3	2	5	25
Dummy	0	0	0	30
Demand	65	30 ₍₁₀₎	50	145/145

❖ Now again, Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7	4	6	70
C	3	2 ₍₁₀₎	5	25 ₍₁₅₎
Dummy	0	0	0	30
Demand	65	30 _{(10) (0)}	50	145/145

❖ Now again, Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7	4	6	70
C	3 ₍₁₅₎	2 ₍₁₀₎	5	25 _{(15) (0)}
Dummy	0	0	0	30
Demand	65 ₍₅₀₎	30 _{(10) (0)}	50	145/145

❖ Now again, Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7	4	6 ₍₅₀₎	70 ₍₂₀₎
C	3 ₍₁₅₎	2 ₍₁₀₎	5	25 _{(15) (0)}
Dummy	0	0	0	30
Demand	65 ₍₅₀₎	30 _{(10) (0)}	50 ₍₀₎	145/145

❖ Now again, Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7 ₍₂₀₎	4	6 ₍₅₀₎	70 _{(20) (0)}
C	3 ₍₁₅₎	2 ₍₁₀₎	5	25 _{(15) (0)}
Dummy	0	0	0	30
Demand	65 _{(50) (30)}	30 _{(10) (0)}	50 ₍₀₎	145/145

❖ Now again, Choose the cell with **least transportation cost**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1 ₍₂₀₎	7	20 ₍₀₎
B	7 ₍₂₀₎	4	6 ₍₅₀₎	70 _{(20) (0)}
C	3 ₍₁₅₎	2 ₍₁₀₎	5	25 _{(15) (0)}
Dummy	0 ₍₃₀₎	0	0	30 ₍₀₎
Demand	65 _{(50) (30) (0)}	30 _{(10) (0)}	50 ₍₀₎	145/145

(c)

⇒ We Know

$$\text{Total Cost} = \Sigma (\text{Allocated Units} \times \text{Cost per Unit})$$

⇒ Total Transportation Cost:

$$\begin{aligned}
 \text{Total Cost} &= (20 \times 1) + (20 \times 7) + (50 \times 6) + (15 \times 3) + (10 \times 2) + (30 \times 0) \\
 &= 20 + 140 + 300 + 45 + 20 + 0 \\
 &= \text{Tk. 525}
 \end{aligned}$$

✂ **Problem 3:** A manufacturing company has three factories located at **A, B, and C**, which supply goods to three warehouses located at **W₁, W₂, W₃**.

The supply available at factories **A, B, and C** is **65 units, 30 units, and 50 units** respectively. The demand at warehouses **W₁, W₂, W₃** is **20 units, 70 units, and 25 units** respectively.

The transportation cost per unit (in Tk.) from each factory to each warehouse is given in the following table:

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1	7	65
B	7	4	6	30
C	3	2	5	50
Demand	20	70	25	

- Determine whether the transportation problem is balanced or unbalanced.
- Obtain a **Basic Feasible Solution (BFS)** using the **Least Cost Method**.
- Calculate the **total transportation cost**.

(a)

Factory \ Warehouse	W ₁	W ₂	W ₃	Supply
A	5	1	7	65
B	7	4	6	30
C	3	2	5	50
Demand	20	70	25	115/145

Since,

$$\sum \text{Supply} \neq \sum \text{Demand}$$

So,

Unbalanced Transportation Problem

(b)

Since **Supply > Demand** → Add a **Dummy Column** to balance the transportation problem.

Factory \ Warehouse	W ₁	W ₂	W ₃	Dummy	Supply
A	5	1	7	0	65
B	7	4	6	0	30
C	3	2	5	0	50
Demand	20	70	25	30	145/145

❖ Choose the cell with **least transportation cost (excluding 0)**:

Factory \ Warehouse	W ₁	W ₂	W ₃	Dummy	Supply
A	5	1 ₍₆₅₎	7	0	65 ₍₀₎
B	7	4	6	0 ₍₃₀₎	30 ₍₀₎
C	3 ₍₂₀₎	2 ₍₅₎	5 ₍₂₅₎	0	50 _{(45) (25) (0)}
Demand	20 ₍₀₎	70 _{(5) (0)}	25 ₍₀₎	30 ₍₀₎	145/145

(c)

⇒ Total Transportation Cost:

$$\begin{aligned}\text{Total Cost} &= (65 \times 1) + (30 \times 0) + (20 \times 3) + (5 \times 2) + (25 \times 5) \\ &= 65 + 0 + 60 + 10 + 125 \\ &= \text{Tk. } 260\end{aligned}$$

Solution (Without Dummy)

⇒ In an unbalanced transportation problem, the solution without dummy ignores the dummy allocations since they have zero cost and do not affect the total transportation cost.

✂ **Problem4:** A company has three sources **Q1, Q2, and Q3** that supply goods to four destinations **D1, D2, D3, and D4**. The supply available at sources Q1, Q2, and Q3 is **300 units, 400 units, and 500 units** respectively. The demand at destinations D1, D2, D3, and D4 is **200 units, 350 units, 400 units, and 200 units** respectively. The transportation cost per unit (in Tk.) from each source to each destination is given in the table below:

Source \ Destination	D1	D2	D3	D4	Supply
Q1	3	1	7	4	300
Q2	2	6	5	9	400
Q3	8	3	3	2	500
Demand	200	350	400	200	

- Determine whether the transportation problem is balanced or unbalanced.
- Obtain a **Basic Feasible Solution (BFS)** using the **Least Cost Method**.
- Calculate the **total transportation cost**.

(a)

Source \ Destination	D1	D2	D3	D4	Supply
Q1	3	1	7	4	300
Q2	2	6	5	9	400
Q3	8	3	3	2	500
Demand	200	350	400	200	1150/1200

Since,

$$\sum \text{Supply} \neq \sum \text{Demand}$$

So,

Unbalanced Transportation Problem

(b)

Source \ Destination	D1	D2	D3	D4	Supply
Q1	3	1 ₍₃₀₀₎	7	4	300 ₍₀₎
Q2	2 ₍₂₀₀₎	6 ₍₅₀₎	5 ₍₁₀₀₎	9	400 _{(200) (100) (50)} (50 unit extra)
Q3	8	3	3 ₍₃₀₀₎	2 ₍₂₀₀₎	500 _{(300) (0)}
Demand	200 ₍₀₎	350 _{(50) (0)}	400 _{(100) (0)}	200 ₍₀₎	1150/1200

❖ **Note:** When equal minimum costs occur in the same row or column, selecting **any one cell is valid and acceptable** but **different choices can lead to different solutions**.

(c)

⇒ We Know

$$\text{Total Cost} = \Sigma (\text{Allocated Units} \times \text{Cost per Unit})$$

⇒ Total Transportation Cost:

$$\begin{aligned}
 \text{Total Cost} &= (300 \times 1) + (200 \times 2) + (50 \times 6) + (100 \times 5) + (300 \times 3) + (200 \times 2) \\
 &= 300 + 400 + 300 + 500 + 900 + 400 \\
 &= \text{Tk. 2800}
 \end{aligned}$$

(b.2)

❖ Choose another **tie** cell to see the difference:

Source \ Destination	D1	D2	D3	D4	Supply
Q1	3	1 ₍₃₀₀₎	7	4	300 ₍₀₎
Q2	2 ₍₂₀₀₎	6	5 ₍₁₅₀₎	9	400 _{(200) (50)} (50 unit extra)
Q3	8	3 ₍₅₀₎	3 ₍₂₅₀₎	2 ₍₂₀₀₎	500 _{(300) (250) (0)}
Demand	200 ₍₀₎	350 _{(50) (0)}	400 ₍₁₅₀₎	200 ₍₀₎	1150/1200

(c.2)

⇒ Total Transportation Cost:

$$\begin{aligned}
 \text{Total Cost} &= (300 \times 1) + (200 \times 2) + (150 \times 5) + (50 \times 3) + (250 \times 3) + (200 \times 2) \\
 &= 300 + 400 + 750 + 150 + 750 + 400 \\
 &= \text{Tk. 2750}
 \end{aligned}$$

- ✂ **Problem5:** A multi-national cloth buyer purchases shirts from three suppliers. The cloth will be supplied to three warehouses. The demand of each warehouse, capacity of each supplier and respected per unit transportation cost (in BDT shown in bold number) are shown in the following table. Using the **Least Cost Method**, identify how many products need to be transported from which supplier(s) to which destination(s). In addition, calculate the total cost associated with it.

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7	9	8	500,000
Supplier-B	6	4	5	400,000
Supplier-C	5	3	4	300,000
Total Demand	500,000	200,000	500,000	

(a)

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7	9	8	500,000
Supplier-B	6	4	5	400,000
Supplier-C	5	3	4	300,000
Total Demand	500,000	200,000	500,000	1200k/1200k

Since,

$$\sum \text{Supply} = \sum \text{Demand}$$

So,

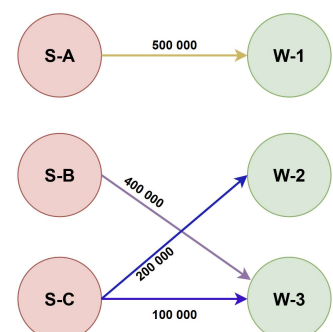
Balanced Transportation Problem

❖ Choose the cell with **least transportation cost**:

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7 (500k)	9	8	500,000(0)
Supplier-B	6	4	5 (400k)	400,000(0)
Supplier-C	5	3 (200k)	4 (100k)	300,000(100k) (0)
Total Demand	500,000(0)	200,000(0)	500,000(400k) (0)	1200k/1200k

⇒ **Total Transportation Cost:**

$$\begin{aligned} \text{Total Cost} &= (500k \times 7) + (400k \times 5) + (200k \times 3) + (100k \times 4) \\ &= \text{Tk. } 6500\ 000 \end{aligned}$$



North-West Corner Method (NWCN)

⇒ The North-West Corner Method starts allocating goods from the **top-left (north-west) cell** of the transportation table and proceeds step by step either **rightward** or **downward**.

❖ Steps of NWCN

1. Start from the **top-left (north-west) cell** of the transportation table.
2. Allocate the **minimum of supply and demand** to that cell.
3. Reduce the supply or demand accordingly.
4. If **supply becomes zero**, move **down** to the next row.
5. If **demand becomes zero**, move **right** to the next column.
6. Repeat until **all supplies and demands are satisfied**.

✂ **Problem6:** A multi-national cloth buyer purchases shirts from three suppliers. The cloth will be supplied to three warehouses. The demand of each warehouse, capacity of each supplier and respected per unit transportation cost (in BDT shown in bold number) are shown in the following table. Using the **North-West Corner Method**, identify how many products need to be transported from which supplier(s) to which destination(s). In addition, calculate the total cost associated with it.

Source	Destinations			
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7	9	8	500,000
Supplier-B	6	4	5	400,000
Supplier-C	5	3	4	300,000
Total Demand	500,000	200,000	500,000	

(a)

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7	9	8	500,000
Supplier-B	6	4	5	400,000
Supplier-C	5	3	4	300,000
Total Demand	500,000	200,000	500,000	1200k/1200k

Since,

$$\sum \text{Supply} = \sum \text{Demand}$$

So,

Balanced Transportation Problem

❖ Start from the **top-left (north-west) cell**:

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7 (500k)	9	8	500,000(0)
Supplier-B	6	4	5	400,000
Supplier-C	5	3	4	300,000
Total Demand	500,000(0)	200,000	500,000	1200k/1200k

❖ Start again from the available **top-left (north-west) cell**:

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7 (500k)	9	8	500,000(0)
Supplier-B	6	4 (200k)	5	400,000(200k)
Supplier-C	5	3	4	300,000
Total Demand	500,000(0)	200,000(0)	500,000	1200k/1200k

❖ Start again from the available **top-left (north-west) cell**:

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7 (500k)	9	8	500,000(0)
Supplier-B	6	4 (200k)	5 (200k)	400,000(200k) (0)
Supplier-C	5	3	4	300,000
Total Demand	500,000(0)	200,000(0)	500,000(300k)	1200k/1200k

❖ Start again from the available **top-left (north-west) cell**:

Source	Destinations			Total Supply
	Warehouse-1	Warehouse-2	Warehouse-3	
Supplier-A	7 (500k)	9	8	500,000(0)
Supplier-B	6	4 (200k)	5 (200k)	400,000(200k) (0)
Supplier-C	5	3	4 (300k)	300,000(0)
Total Demand	500,000(0)	200,000(0)	500,000(300k) (0)	1200k/1200k

⇒ **Total Transportation Cost:**

$$\begin{aligned}
 \text{Total Cost} &= (500k \times 7) + (200k \times 4) + (200k \times 5) + (300k \times 4) \\
 &= \text{Tk. } 6500\ 000
 \end{aligned}$$

INVENTORY MANAGEMENT

Inventory Management

⇒ Inventory management is a **core operations management activity**

➤ **It affects:**

- Operations
- Marketing
- Finance

➤ **Poor inventory management causes:**

- Stockouts or excess inventory
- Higher operating cost
- Lower customer satisfaction

➤ **Good inventory management ensures:**

- Smooth production
- Better service level
- Cost efficiency

Introduction to Inventory

- **Inventory:** Stock or store of goods
- Firms carry many items ranging from small tools to large machines
- Inventory depends on business type:
 - **Manufacturing:** raw materials, WIP, finished goods
 - **Retail:** clothing, food, stationery
 - **Hospitals:** medicines, equipment
- Inventory investment:
 - ~30% of current assets
 - Up to 90% of working capital

Types of Inventories

1. Raw materials and purchased parts
2. Work-in-Process (WIP)
3. Finished goods / merchandise
4. Tools and supplies
5. MRO inventory (Maintenance, Repair, Operations)
6. Pipeline inventory (goods in transit)

Note: Space limitation can affect inventory decisions.

Inventory Based on Purpose

1. **Cycle Stock:** Inventory to meet regular demand
2. **Safety Stock:** Extra inventory to prevent stockouts
3. **Anticipation Stock:** Inventory held for expected future demand
4. **Pipeline Inventory:** Inventory moving through supply chain

Why We Need Inventory

⇒ Inventory helps to:

- Continue production during supply delays
- Support production when machines fail
- Meet high market demand
- Ensure continuous supply even with disruptions



Inventory Counting Systems

⇒ Two main systems:

1. **Periodic Inventory System**
2. **Perpetual (Continuous Review) Inventory System**



- Inventory counted at fixed intervals (weekly/monthly)
- Order placed after physical count
- **Advantages:**
 - Simple
 - Orders placed together
- **Disadvantages:**
 - No control between reviews
 - Higher safety stock required

❖ **Example:** Ordering fish every Thursday for Friday delivery

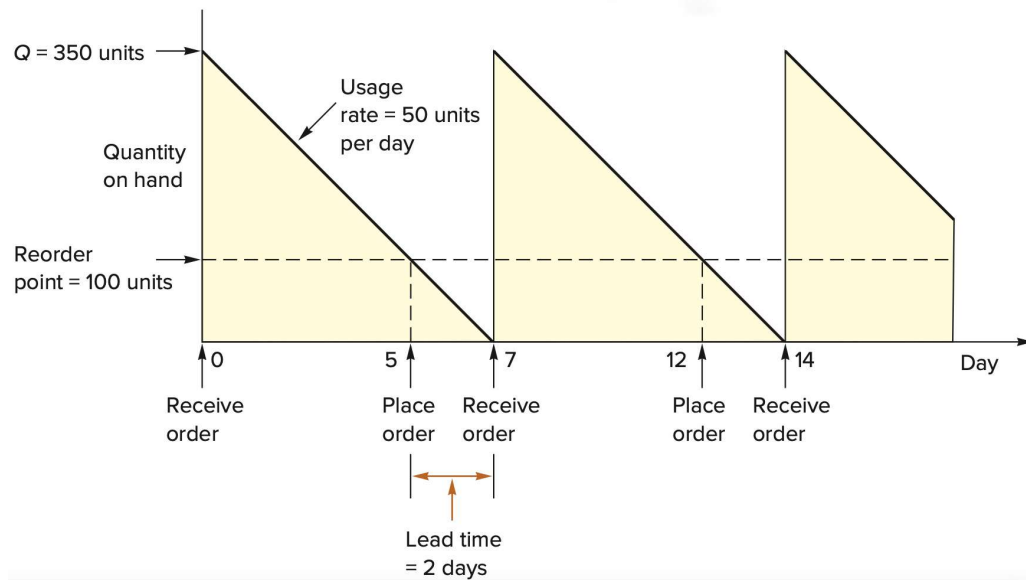
Perpetual Inventory System

- Inventory updated continuously
- Order placed when stock reaches a minimum level
- **Advantages:**
 - Better control
 - Optimal order quantity possible
- **Disadvantages:**
 - Higher record-keeping cost
 - Physical verification still required

❖ **Example:** Order when inventory reaches 10 units

Inventory Cycle

- Inventory decreases at a constant rate
- Order placed at **Reorder Point (ROP)**
- New stock arrives after **Lead Time**



- Order size, $Q = 350$ units
- Usage rate = 50 units per day
- Lead time = 2 days
- Reorder point = 100 units (2 days' supply)

When to Order: Reorder Point (ROP)

- Indicates **when to place an order**
- Mostly used in perpetual systems

➤ Formula:

$$\text{ROP} = d \times \text{LT}$$

Where:

- d = demand per day
- LT = lead time

✂ **Problem1:** A retail store experiences a constant daily demand of **32.154 units** for a particular product. The supplier requires a **lead time of 10 days** to deliver the order.

a) Calculate the **Reorder Point (ROP)** for the product.

(a)

⇒ **Is given,**

$$d = 32.154 \text{ Unit}$$

$$LT = 10 \text{ days}$$

⇒ **We Know**

$$\text{ROP} = d \times LT$$

$$= 32.154 \times 10 = 321.540 \text{ units}$$

✂ **Problem2:** The demand for a product is **10,000 yards per year**, and the store remains open **311 days annually**. The supplier's lead time is **10 days**.

Assuming demand and lead time are constant:

- Calculate the daily demand
- Compute the reorder point
- Identify which inventory system (periodic or perpetual) is most appropriate for using ROP

(a)

⇒ **Is given,**

$$\text{Demand}(D) = 10,000 \text{ yards/year}$$

$$LT = 10 \text{ days}$$

$$\text{Store Open} = 311 \text{ days}$$

⇒ **We Know**

$$d = \frac{D}{\text{Working Days}}$$

$$= \frac{10,000}{311} = 32.1540 \text{ yards/day}$$

(b)

⇒ **Is given,**

$d = 32.154 \text{ yards/day}$

$LT = 10 \text{ days}$

⇒ **We Know**

$$ROP = d \times LT$$

$$= 32.154 \times 10 = 321.540 \text{ yards}$$

(c)

➔ The **Perpetual (Continuous Review) Inventory System** is most appropriate for using ROP because:

- Inventory levels are monitored continuously.
- An order is placed exactly when inventory reaches the reorder point.
- This minimizes the risk of stockouts and excess inventory.

Inventory Costs

➤ Four basic costs:

1. **Purchase Cost**
2. **Holding (Carrying) Cost**
3. **Ordering Cost**
4. **Shortage Cost**

Purchase Cost

- Cost paid to supplier
- Includes product price and shipping
- Largest inventory cost

Holding (Carrying) Cost

➤ Includes:

- Storage, insurance, taxes
- Obsolescence, spoilage, damage
- Security and handling
- Opportunity cost of invested capital

Formula:

$$THC = \frac{Q}{2} \times H$$

Q = Order Quantity

H = Holding (Carrying) Cost per Unit per Year

Ordering Cost

- Cost of placing and receiving orders
- Includes paperwork, inspection, transport
- **Fixed per order**
- **In production: setup cost**

Formula:

$$TOC = \frac{D}{Q} \times S$$

Q = Order Quantity

S = Ordering Cost per Order

D = Annual Demand

Shortage Cost

Occurs when demand > supply

➤ Includes:

- Lost sales
- Loss of goodwill
- Backorder and penalty costs

Economic Order Quantity (EOQ)

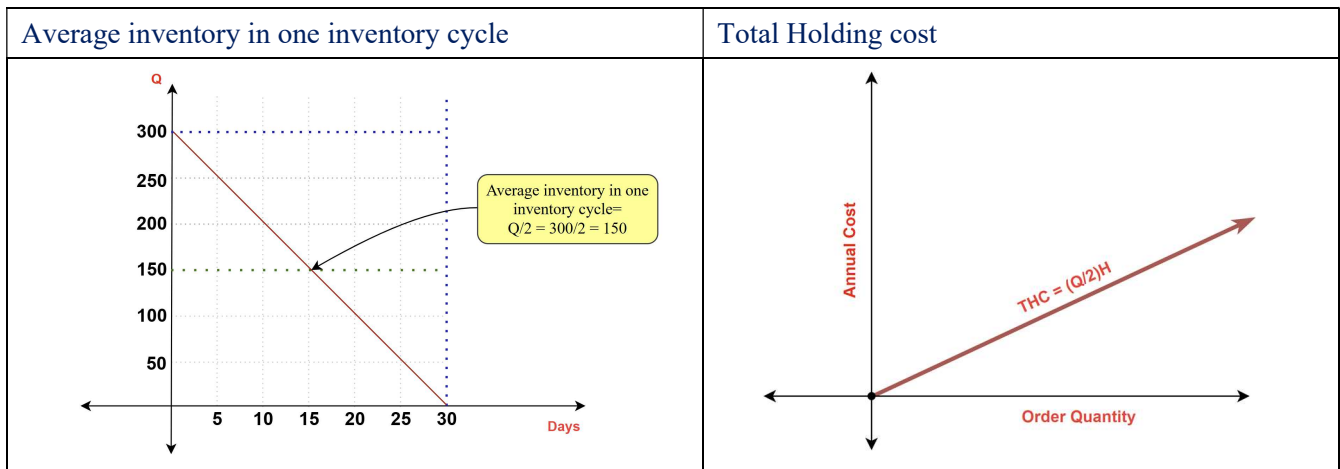
- **Determines optimal order quantity**
- Minimizes:
 - Ordering cost
 - Holding cost
- **Answers: How much to order**
- **Does NOT answer: When to order**

EOQ Assumptions

1. Constant demand
2. Constant lead time
3. No shortages
4. Instant replenishment

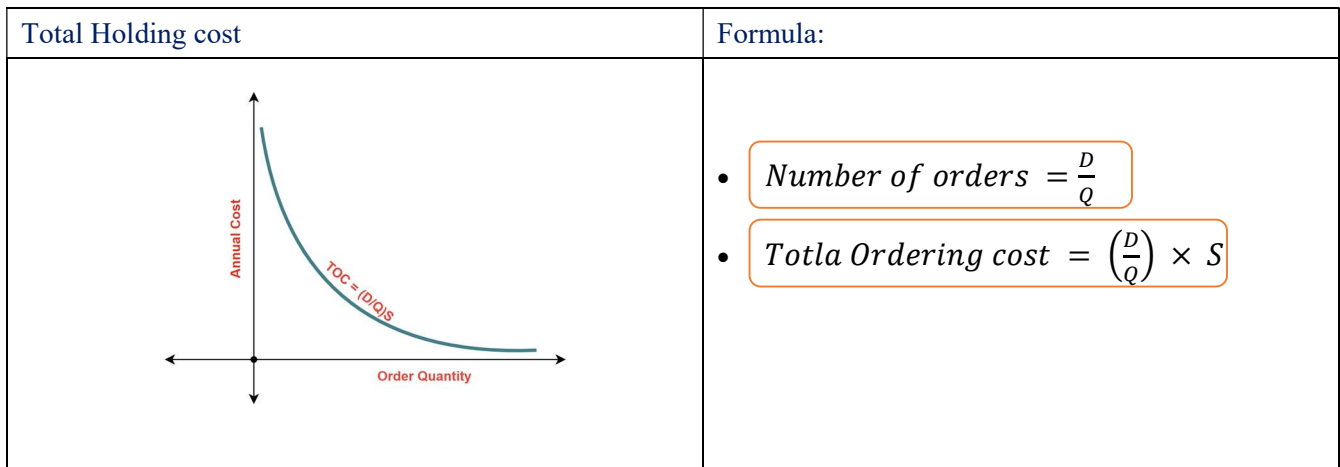
EOQ Cost Components

❖ Holding Cost

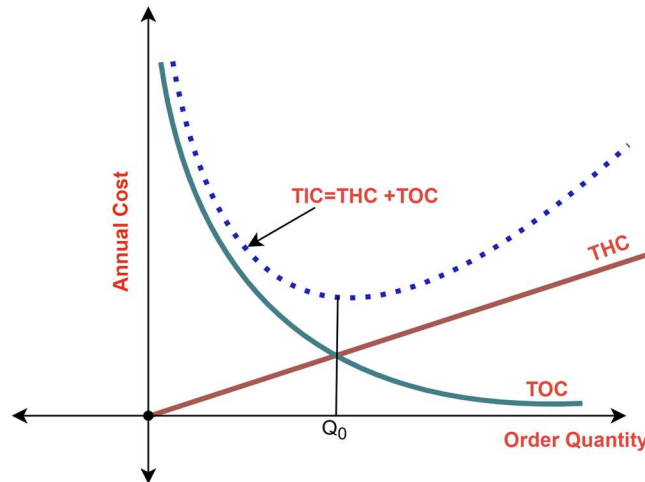


- $Average\ inventory = \frac{Q}{2}$
- $Total\ Holding\ cost = \left(\frac{Q}{2}\right) \times H$

❖ Ordering Cost



Total Inventory Cost



$$TIC = \text{Total Holding Cost} + \text{Total Ordering Cost}$$

$$TIC = \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S$$

- TC curve is **U-shaped**
- Minimum cost occurs where:
 - **Holding cost = Ordering cost**

EOQ Formula

Holding cost = Ordering cost

$$\left(\frac{Q}{2}\right) \times H = \left(\frac{D}{Q}\right) \times S$$

$$Q^2 H = 2DS$$

$$Q_0 \text{ or EOQ} = \sqrt{\frac{2DS}{H}}$$

Where:

- D = annual demand
- S = ordering cost per order
- H = holding cost per unit per year

Order Cycle Length

⇒ Time between two orders

$$\text{Order Cycle} = \frac{\text{EOQ}}{D} \times \text{Firm Working Days}$$

✂ **Problem3:** A company has an annual demand of **10,000 yards** for a particular product. The ordering cost is **\$150 per order**, and the annual holding cost is **\$0.75 per yard**. The supplier's lead time is **7 days**. Assume the firm operates **365 days per year** and demand is constant.

- a) Determine the **Economic Order Quantity (EOQ)**.
- b) Calculate the **minimum total annual inventory cost**
- c) Find the **optimum number of orders per year**.
- d) Determine the **length of the order cycle**.
- e) Calculate the **reorder point (ROP)**.

(a)

⇒ **Is given,**

Demand(D) = 10,000 yards/year

LT = 7 days

firm operates = 365 days

S = \$150 per order

H = \$0.75 per year

⇒ **We Know**

$$\begin{aligned}\text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 10,000 \times 150}{0.75}} = 2000 \text{ yards}\end{aligned}$$

(b)

⇒ **We Know**

$$\begin{aligned}\text{TIC} &= \left(\frac{\text{EOQ}}{2}\right) \times H + \left(\frac{D}{\text{EOQ}}\right) \times S \\ &= \left(\frac{2,000}{2}\right) \times 0.75 + \left(\frac{10,000}{2,000}\right) \times 150 = \$1500 \text{ per year}\end{aligned}$$

(c)

⇒ We Know

$$\begin{aligned}\text{Number of orders} &= \left(\frac{D}{EOQ} \right) \\ &= \left(\frac{10,000}{2000} \right) = 5 \text{ order/year}\end{aligned}$$

(d)

⇒ We Know

$$\begin{aligned}\text{Order Cycle} &= \frac{EOQ}{D} \times \text{Firm Working Days} \\ &= \left(\frac{2000}{10,000} \right) \times 365 = 73 \text{ day}\end{aligned}$$

(e)

⇒ We Know

$$\begin{aligned}d &= \frac{D}{\text{Working Days}} \\ &= \frac{10,000}{365} = 27.397 \text{ yards/day}\end{aligned}$$

$$\text{ROP} = d \times \text{LT}$$

$$= 27.397 \times 7 = 191.781 \text{ yards}$$

✂ **Problem4:** A local distributor for a national tire company expects to sell approximately **9,600 steel-belted radial tires** of a certain size and tread design next year. The **annual carrying cost** is **\$16 per tire**, and the **ordering cost** is **\$75 per order**. The distributor operates **288 days a year**.

- What is the **Economic Order Quantity (EOQ)**?
- How many times per year does the store reorder?
- What is the length of an order cycle?
- What will the total annual cost be if the EOQ quantity is ordered?

(a)

⇒ Is given,

Demand(D) = 9600 tires /year

operates = 288 days/year

S = \$75 per order

H = \$16 per year

⇒ We Know

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 9600 \times 75}{16}} = 300 \text{ tires} \end{aligned}$$

(b)

⇒ We Know

$$\begin{aligned} \text{Number of orders} &= \left(\frac{D}{\text{EOQ}} \right) \\ &= \left(\frac{9,600}{300} \right) = 32 \text{ order/year} \end{aligned}$$

(c)

⇒ We Know

$$\begin{aligned} \text{Order Cycle} &= \frac{\text{EOQ}}{D} \times \text{Firm Working Days} \\ &= \left(\frac{300}{9600} \right) \times 288 = 9 \text{ day} \end{aligned}$$

(d)

⇒ We Know

$$\begin{aligned} \text{TIC} &= \left(\frac{\text{EOQ}}{2} \right) \times H + \left(\frac{D}{\text{EOQ}} \right) \times S \\ &= \left(\frac{300}{2} \right) \times 16 + \left(\frac{9600}{300} \right) \times 75 = \$4800 \text{ per year} \end{aligned}$$

✂ **Problem5:** Piddling Manufacturing assembles security systems. It purchases **3,600 high-definition security cameras per year** at **\$180 each**. **Ordering costs** are **\$50 per order**, and **annual carrying costs** are **20 percent of the purchase price**.

- a. What is the **Economic Order Quantity (EOQ)**?
- b. What will the total annual cost be if the EOQ quantity is ordered?

(a)

⇒ **Is given,**

Demand(D) = 3600 cameras /year

$S = \$50$ per order

$H = \$180 \times 20\% = 36$ per year

⇒ **We Know**

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 3600 \times 50}{36}} = 100 \text{ cameras} \end{aligned}$$

(b)

⇒ **We Know**

$$\begin{aligned} \text{TIC} &= \left(\frac{\text{EOQ}}{2} \right) \times H + \left(\frac{D}{\text{EOQ}} \right) \times S \\ &= \left(\frac{100}{2} \right) \times 36 + \left(\frac{3600}{100} \right) \times 50 = \$3600 \text{ per year} \end{aligned}$$

✂ **Problem6:** One of the top-selling items in the container group at a museum's gift shop is a **bird feeder**. The average sales rate is **18 units per week**, and the supplier charges **Tk. 60 per unit**. The cost of placing an order is **Tk. 45 per order**. The **annual holding cost** is **25% of the unit purchase price**. The museum operates **52 weeks per year**. Currently, management orders **390 units per order** so that orders are placed less frequently.

- a) Determine the **Economic Order Quantity (EOQ)**.
- b) Calculate the **annual inventory cost** for the current order size of **390 units**.
- c) Calculate the **annual inventory cost** when ordering the **EOQ**.
- d) The supplier offers a **10% discount** on the unit price if **500 units** are ordered at a time. Considering all relevant costs, determine whether the museum should **accept or reject the discount offer**. Justify your answer.

(a)

⇒ Is given,

avg selling unit = 18 unit /week

Purchase cost = 60 tk /unit

$S = 45\text{tk/order}$

$H = 60 \times 25\% = 15\text{tk/year}$

operates = 52 weeks/year

Order size, $Q = 390$ unit /order

$\therefore D = \text{Average demand per week} \times \text{Number of weeks per year}$

$$\therefore D = 18 \times 52 = 936 \text{ unit/year}$$

⇒ We Know

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 936 \times 45}{15}} \approx 75 \text{ units} \end{aligned}$$

(b)

⇒ We Know

$$\begin{aligned} \text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{390}{2}\right) \times 15 + \left(\frac{936}{390}\right) \times 45 = 3033\text{tk per year} \end{aligned}$$

(c)

⇒ We Know

$$\begin{aligned} \text{TIC} &= \left(\frac{\text{EOQ}}{2}\right) \times H + \left(\frac{D}{\text{EOQ}}\right) \times S \\ &= \left(\frac{75}{2}\right) \times 15 + \left(\frac{936}{75}\right) \times 45 = 1124.10\text{tk per year} \end{aligned}$$

(d)

⇒ Is given,

Purchase cost = $60 - 60 \times 10\% = 54\text{tk /unit}$

$S = 45\text{tk/order}$

$H = 54 \times 25\% = 13.5\text{tk/year}$

$D = 936 \text{ unit /year}$

Order size, $Q = 500 \text{ unit /order}$

⇒ We Know

$$\begin{aligned}\text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{500}{2}\right) \times 13.5 + \left(\frac{936}{500}\right) \times 45 = 3459.24\text{tk per year}\end{aligned}$$

Total Annual Cost = TIC + Total Purchase cost

$$\text{TAC}_{\text{with discount}} = 3459.24 + 54 \times 936 = 54003.24\text{tk/year}$$

⇒ For EOQ

Purchase cost = 60 tk /unit

$D = 936 \text{ unit /year}$

$\text{TIC} = 1124.10 \text{ tk /year}$

⇒ We Know

$$\text{TAC}_{\text{EOQ}} = 1124.10 + 60 \times 936 = 57284.10\text{tk/year}$$

Since,

$$\text{TAC}_{\text{with discount}} < \text{TAC}_{\text{EOQ}}$$

So, the museum should Accept the supplier's discount offer and order 500 units

✂ **Problem7:** Two **RMG (Readymade Garments)** companies purchase cotton from China. The **monthly demand** for cotton for each company is **300 tons**.

- **Company-1** purchases the **entire monthly requirement at the beginning of the month** in a single order.
- **Company-2** purchases cotton in **three equal orders of 100 tons each** during the month.

For both companies:

- **Ordering cost = \$1,000 per order**
- **Holding cost = \$30 per ton per month.**

Assume demand is constant and shortages are not allowed.

- a) Calculate the **total monthly inventory cost** for **Company-1**.
- b) Calculate the **total monthly inventory cost** for **Company-2**.
- c) Identify **which company's purchasing strategy is more cost-effective** and justify your answer.
- d) Determine the **Economic Order Quantity (EOQ)** for cotton.
- e) Based on the EOQ result, **suggest a better purchasing policy** for both companies.

(a)

⇒ **Is given for C-1,**

Demand(D) = 300 tons /month

No of order = 1

$S = \$1000$ per order

$H = \$30$ per month

$Q = 300$ tons/Orders

⇒ **We Know**

$$\begin{aligned} \text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{300}{2}\right) \times 30 + \left(\frac{300}{300}\right) \times 1000 = \$5500 \text{ per month} \end{aligned}$$

(b)

⇒ **Is given for C-2,**

Demand(D) = 300 tons /month

No of order = 3

$S = \$1000$ per order

$H = \$30$ per month

$Q = 100$ tons/Orders

⇒ **We Know**

$$\begin{aligned}\text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{100}{2}\right) \times 30 + \left(\frac{300}{100}\right) \times 1000 = \$4,500 \text{ per month}\end{aligned}$$

(c)

⇒ **We have,**

$\text{TIC}_{C-1} = \$5500$ per month

$\text{TIC}_{C-2} = \$4500$ per month

Since,

$$\text{TAC}_{C-2} < \text{TAC}_{C-1}$$

So, Company-2's strategy is better, as it saves: $5,500 - 4,500 = \$1,000$ per month

(d)

⇒ **Is given,**

Demand(D) = 300 tons /month

$S = \$1000$ per order

$H = \$30$ per month

⇒ **We Know**

$$\begin{aligned}\text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 300 \times 1000}{30}} = 141.421 \text{ tons}\end{aligned}$$

(e)

- Order approximately **141.421 tons per order**
- Number of orders per month:

$$\frac{D}{EOQ} = \frac{300}{141.421} = 2.121 \text{ orders/month}$$

❖ This EOQ-based policy:

- Minimizes total inventory cost
- Balances ordering and holding costs
- Is **better than both Company-1 and Company-2 strategies**

✂ **Problem8:** Two pharmaceutical companies purchase packaging bottles from a local supplier. The **monthly demand** for each company is **200 units**.

- **Company-A** places **two equal order of 100 units** during the month.
- **Company-B** places **four equal orders of 50 units** during the month.

For both companies:

- **Ordering cost = \$500 per order**
- **Holding cost = \$24 per unit per year**

- a) Calculate the **total monthly inventory cost** for **Company-A**.
- b) Calculate the **total monthly inventory cost** for **Company-B**.
- c) Identify **which company's strategy is more cost-effective**.
- d) Calculate the **Economic Order Quantity (EOQ)**.
- e) Suggest a **better ordering policy** based on EOQ.

(a)

⇒ **Is given for C-A,**

Demand(D) = 200 unit /month

No of order = 2

S = \$500 per order

$H = \frac{\$24}{12} = \2 per month

Q = 100 units/Orders

⇒ We Know

$$\begin{aligned}\text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{100}{2}\right) \times 2 + \left(\frac{200}{100}\right) \times 500 = \$1,100 \text{ per month}\end{aligned}$$

(b)

⇒ Is given for C-B,

Demand(D) = 200 unit /month

No of order = 4

$S = \$500$ per order

$H = \frac{\$24}{12} = \2 per month

$Q = 50$ unit/Orders

⇒ We Know

$$\begin{aligned}\text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{50}{2}\right) \times 2 + \left(\frac{200}{50}\right) \times 500 = \$2050 \text{ per month}\end{aligned}$$

(c)

⇒ We have,

$\text{TIC}_{C-A} = \$1100$ per month

$\text{TIC}_{C-B} = \$2050$ per month

Since,

$$\text{TAC}_{C-A} < \text{TAC}_{C-B}$$

So, Company-A's strategy is better, as it saves: $2,050 - 1,100 = \$950$ per month

(d)

⇒ Is given,

Demand(D) = 200 units /month

$S = \$500$ per order

$H = \frac{\$24}{12} = \2 per month

⇒ We Know

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 200 \times 500}{2}} \approx 316 \text{ units} \end{aligned}$$

(e)

- Order approximately **316 units per order**
- Number of orders per month:

$$\frac{D}{\text{EOQ}} = \frac{200}{316} = 0.633 \text{ orders/month}$$

❖ This EOQ-based policy:

- Place **one order roughly every 1.5 months**
- This EOQ-based policy minimizes total inventory cost and is **better than both Company-A and Company-B strategies**

✂ Problem9: This case study is carried out in company called as XYZ. This company was established in the year 2008, the company manufacture and supply different types of blowers. The Company has supplied their products to many customers in India. The Company has office in Nallasopara (E.), Maharashtra for marketing purpose. Raw material required for manufacturing blowers are H.R. Sheet, Motors of different H.P., C-channel, L-angle, nut-bolt and paint. The Main issue of this industry is inventory control for ordering raw material and supply of blowers to customers. To manage inventory of raw materials a case study is carried out and EOQ method of inventory management has been adopted. Current forecasting method used by this firm has brought some problem due to inaccurate forecasting. Forecasting method used is the simple average based on the average historical demand, but this has led to inaccurate prediction. The purpose of this study is to recommend alternatives ways to help the company

to reduce the stock and cost by using more effective prediction EOQ and ROP. For this purpose, analysis of a product of the company is done, using data of the last one year.

Annual demand for lug is calculated based on average monthly turnover in this firm. The demand for H.R. Sheet is 3000 kg per month. Cost of one unit is 60 Tk per kg. According to Company's current forecasting model, **company makes order one time in a month** (current lot size) and the ordering cost is 20,000Tk for each time the order is made. The H.R. Sheets are stored in a warehouse that costs 30 Tk to store 1 kg H.R. Sheets for a month.

Based on the available information,

- Suggest the lot size at which the H.R. Sheets should be purchased that will make the total inventory cost minimum.
- How much money the company can save by procuring steel at a new lot size.
- Draw the inventory cycle for four months in both cases- current lot size and EOQ
- If the lead time of the supplier is 1 week, what will be the reorder point in both scenarios.

(a)

⇒ **Is given,**

Demand(D) = 3000 kg /month

$S = 20,000 \text{ TK/order}$

$H = 30 \text{ TK/month}$

$Q = 3000 \text{ kg/Orders}$

⇒ **We Know**

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 3000 \times 20,000}{30}} = 2000 \text{ kg} \end{aligned}$$

(b)

⇒ Is given for initial plan,

Demand(D) = 3000 kg /month

$S = 20,000 \text{ TK}$

$H = 30 \text{ TK}$

$Q = 3000 \text{ kg/Orders}$

⇒ We Know

$$\begin{aligned} \text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{3000}{2}\right) \times 30 + \left(\frac{3000}{3000}\right) \times 20,000 = 65000 \text{ tk per month} \end{aligned}$$

⇒ Is given for optimal plan,

Demand(D) = 3000 kg /month

$S = 20,000 \text{ TK}$

$H = 30 \text{ TK}$

$Q = 2000 \text{ kg/Orders}$

⇒ We Know

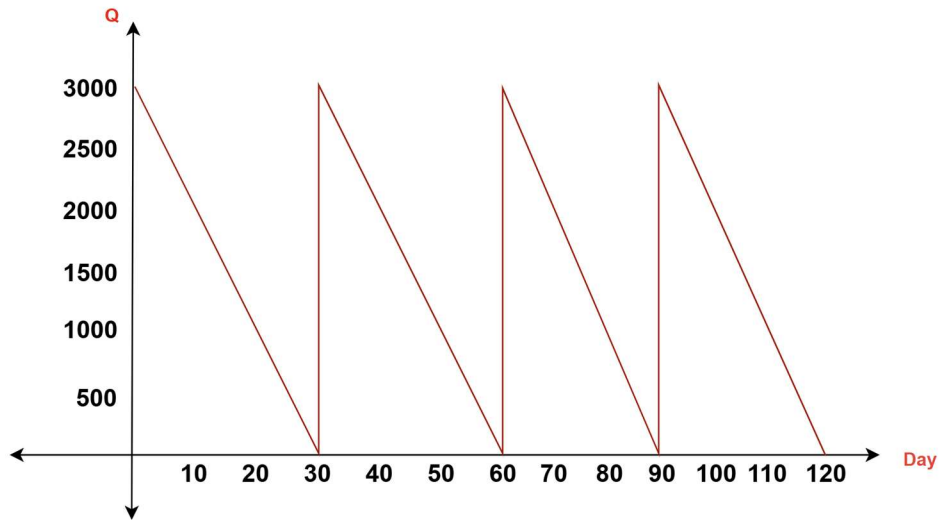
$$\begin{aligned} \text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{2000}{2}\right) \times 30 + \left(\frac{3000}{2000}\right) \times 20,000 = 60000 \text{ tk per month} \end{aligned}$$

saves: $65,000 - 60,000 = 5,000 \text{ tk per month}$

(c)

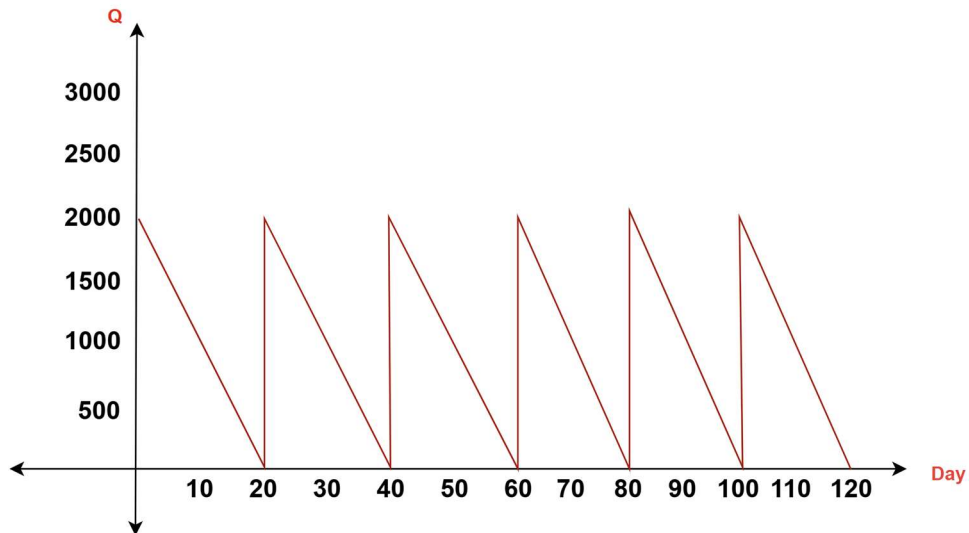
⇒ For initial plan,

$$\begin{aligned}\text{Order Cycle} &= \frac{Q}{D} \times \text{Firm Working Days} \\ &= \frac{3000}{3000} \times 30 = 30 \text{ day}\end{aligned}$$



⇒ For optimal plan,

$$\begin{aligned}\text{Order Cycle} &= \frac{EOQ}{D} \times \text{Firm Working Days} \\ &= \frac{2000}{3000} \times 30 = 20 \text{ day}\end{aligned}$$



(d)

⇒ Is given for initial plan,

$$LT = 1 \text{ Week} = 7 \text{ days}$$

$$d = \frac{3000}{30} = 100 \text{ kg/day}$$

⇒ We Know

$$ROP = d \times LT$$

$$= 100 \times 7 = 700 \text{ kg}$$

⇒ Is given for optimal plan,

$$LT = 1 \text{ Week} = 7 \text{ days}$$

$$d = \frac{2000}{20} = 100 \text{ kg/day}$$

⇒ We Know

$$ROP = d \times LT$$

$$= 100 \times 7 = 700 \text{ kg}$$

✂ **Problem10:** Annual demand for **packaged drinking water bottles** is estimated based on the average monthly sales of a distribution company. The demand for drinking water bottles is **4000 units per month**. The cost of one unit is **Tk 60 per bottle**. According to the company's current purchasing policy, the company places an order **once every month**, and the **ordering cost is Tk 10,000 for each order placed**. The drinking water bottles are stored in a warehouse, and the **inventory holding cost is Tk 240 per unit per year**. The company operates **48 weeks per year** (4 weeks per month). At present, management orders **4000 units per order** so that orders are placed less frequently.

- Suggest the **lot size** at which the drinking water bottles should be purchased so that the **total inventory cost is minimized**.
- Determine **how much money the company can save annually** by procuring drinking water bottles at the new lot size instead of the current lot size.
- If the lead time of the supplier is 2 week, what will be the **reorder point** in both scenarios.
- Draw the **inventory cycle** for three months in both cases- current lot size and EOQ. Identify the **average inventory and re-ordering point** in each diagram.

(a)

⇒ Is given,

Demand(D) = 4000 unit /month

$S = 10,000 \text{ TK/order}$

$H = 240 \text{ TK/year} = 240/48 = 5 \text{ TK/weeks} = 5 \times 4 = 20 \text{ TK/month}$

$Q = 4000 \text{ unit/Orders}$

⇒ We Know

$$\begin{aligned} \text{EOQ} &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 4000 \times 10,000}{20}} = 2000 \text{ unit} \end{aligned}$$

(b)

⇒ Is given for initial plan,

Demand(D) = 4000 unit /month

$S = 10,000 \text{ TK/order}$

$H = 20 \text{ TK/month}$

$Q = 4000 \text{ unit/Orders}$

⇒ We Know

$$\begin{aligned} \text{TIC} &= \left(\frac{Q}{2}\right) \times H + \left(\frac{D}{Q}\right) \times S \\ &= \left(\frac{4000}{2}\right) \times 20 + \left(\frac{4000}{4000}\right) \times 10,000 = 50000 \text{ tk/month} \end{aligned}$$

⇒ Is given for optimal plan,

Demand(D) = 4000 unit /month

$S = 10,000 \text{ TK/order}$

$H = 20 \text{ TK/month}$

$Q_0 = 2000 \text{ unit/Orders}$

⇒ We Know

$$\begin{aligned}\text{TIC} &= \left(\frac{Q_0}{2}\right) \times H + \left(\frac{D}{Q_0}\right) \times S \\ &= \left(\frac{2000}{2}\right) \times 20 + \left(\frac{4000}{2000}\right) \times 10,000 = 40000 \text{tk/month}\end{aligned}$$

∴ saves: $50,000 - 40,000 = 10,000 \text{tk per month}$

∴ saves: $10,000 \times 12 = 120,000 \text{ tk per year}$

(c)

⇒ Is given,

$$LT = 2 \text{ Week} = 14 \text{ day}$$

$$d = \frac{4000}{4} = 1000 \text{ unit/week} = 1000/7 = 142.857 \text{ per day}$$

⇒ We Know

$$\text{ROP} = d \times LT$$

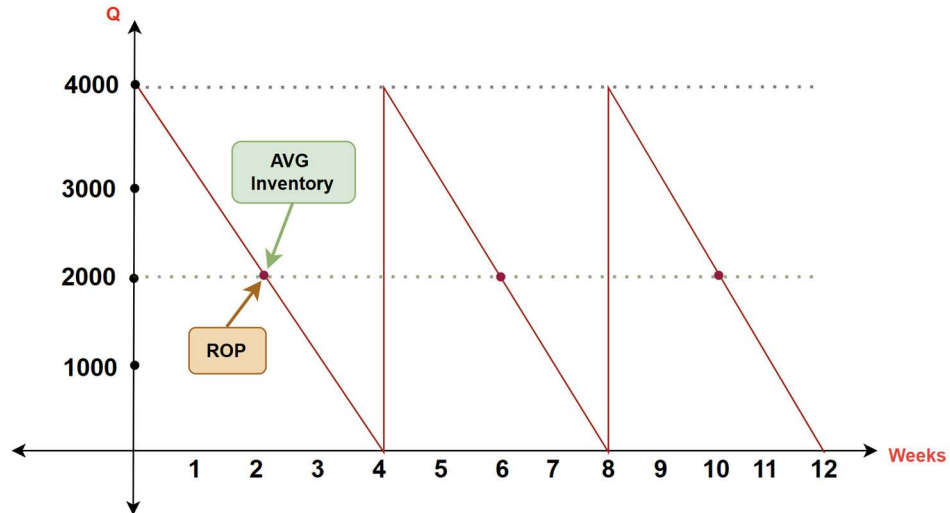
$$= 142.857 \times 14 = 2000 \text{ unit}$$

∴ ROP For both case = 2000 unit

(d)

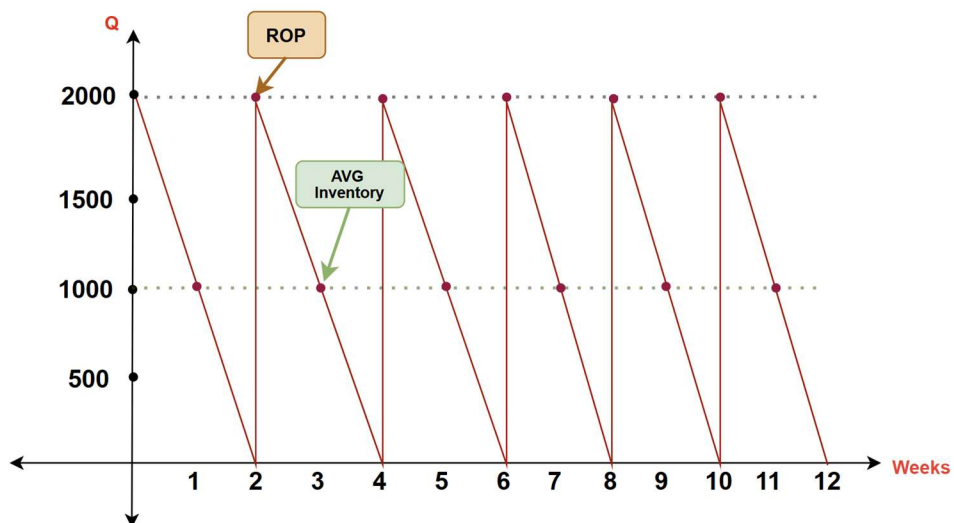
⇒ For initial plan,

$$\begin{aligned}\text{Order Cycle} &= \frac{Q}{D} \times \text{Firm Working Days} \\ &= \frac{4000}{4000} \times 4 = 4 \text{ Weeks}\end{aligned}$$



⇒ For optimal plan,

$$\begin{aligned}\text{Order Cycle} &= \frac{EOQ}{D} \times \text{Firm Working Days} \\ &= \frac{2000}{4000} \times 4 = 2 \text{ Weeks}\end{aligned}$$



PROJECT MANAGEMENT

Project Management – Introduction

⇒ **Project:** A project is a **unique, one-time activity** designed to achieve specific objectives within a **limited time frame**.

❖ Examples:

- Constructing a shopping complex
- Merging companies
- Political campaigns
- Software development
- Database and web design

Characteristics of a Project

- Has **fixed and predefined objectives**
- Has a **limited timeframe**
- Has **limited budget and resources**
- **Uniqueness:** Even similar projects differ due to location, resources, and environment

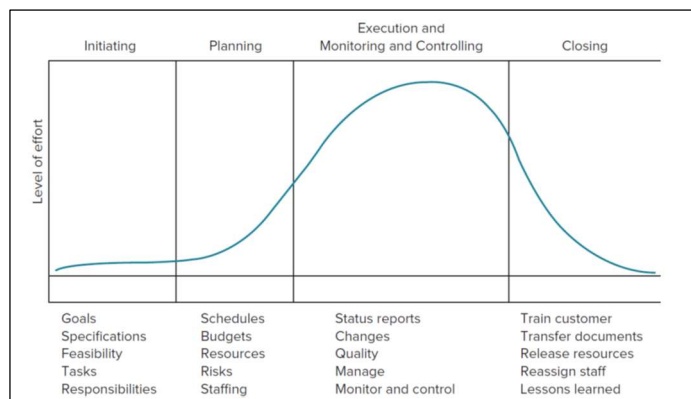
Project vs Operations

Project	Operations
Temporary	Continuous
One-time	Repetitive
Goal-oriented	Efficiency-oriented
Example: Product design	Example: Manufacturing

Project Life Cycle

⇒ Every project follows a **life cycle** consisting of **five phases**:

1. Initiating
2. Planning
3. Executing
4. Monitoring & Controlling
5. Closing



Initiating Phase	Planning Phase
• Defines costs, benefits, and risks • Identifies project goals • Selects the project manager	• Defines: ◦ Project scope ◦ Deliverables ◦ Budget ◦ Schedule and milestones • Identifies: ◦ Required resources ◦ Quality plan ◦ Risk management plan • Selects project team members
Executing Phase	Monitoring and Controlling Phase
• Actual project work is performed • Resources are used and milestones are achieved • Managed using nine PM areas: 1. Integration 2. Scope 3. Human Resources 4. Communication 5. Time 6. Risk 7. Quality 8. Cost 9. Procurement	• Runs parallel with execution • Compares actual progress vs planned progress • Takes corrective actions if needed
Closing Phase	
• Project is formally completed • Deliverables are handed over • Customer acceptance is obtained • Lessons learned are documented • Resources are released	

Key Decisions in Project Management

- Selecting projects
- Selecting project manager
- Selecting project team
- Planning and designing project
- Managing project resources
- Deciding project termination

The Project Manager

⇒ The project manager is **responsible for project success or failure**.

❖ Responsibilities:

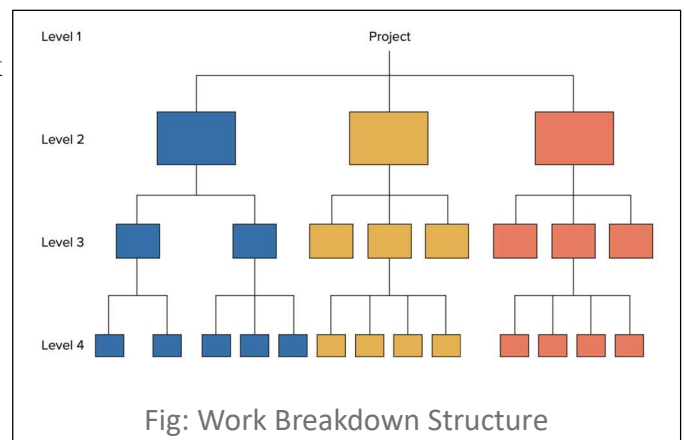
1. Managing work and activities
2. Managing human resources
3. Ensuring effective communication
4. Maintaining quality
5. Completing project on time
6. Completing project within budget
7. Preventing **scope creep**

Project Management Tools

- Work Breakdown Structure (WBS)
- Gantt Chart
- PERT and CPM

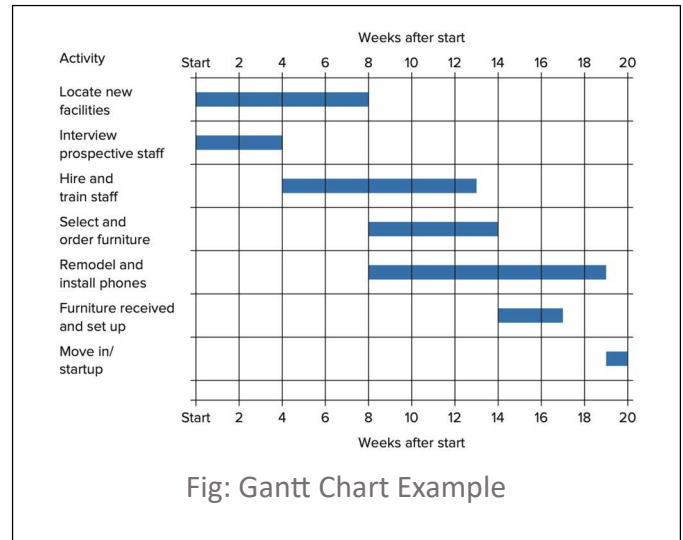
Work Breakdown Structure (WBS)

- A **hierarchical breakdown** of project work
- Identifies **all activities** required
- Levels:
 - Level 2: Major elements
 - Level 3: Supporting activities
 - Level 4: Detailed tasks
- Large projects may have more levels



Gantt Chart

- A **visual scheduling tool**
- Shows:
 - Activities
 - Duration
 - Start and finish times
- Used to track **planned vs actual progress**
- Best for simple projects



PERT and CPM

- **PERT**: Program Evaluation and Review Technique
- **CPM**: Critical Path Method

❖ Advantages:

- Graphical view of activities
- Estimates project duration
- Identifies critical activities
- Determines allowable delays

Critical Path Analysis (CPA)

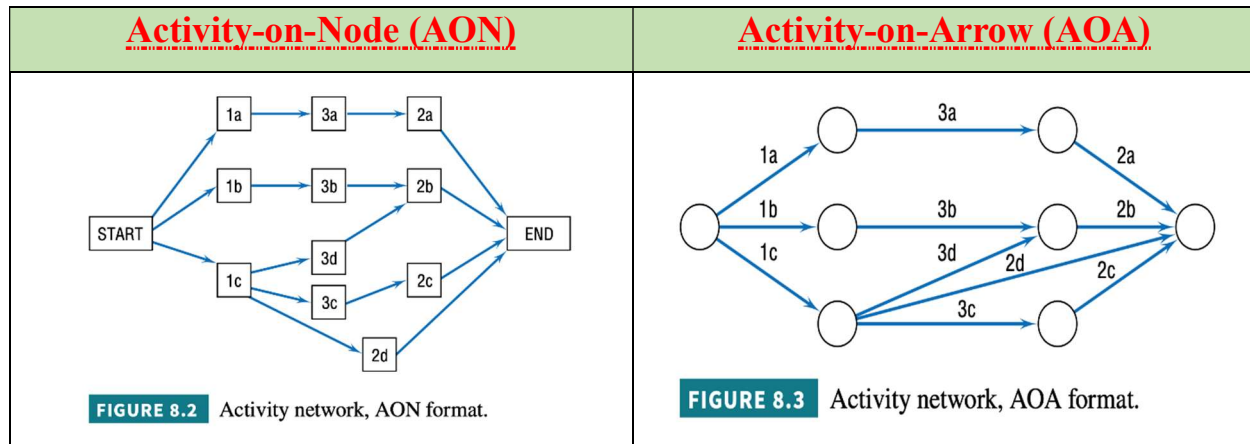
- ⇒ Identifies the **longest path** in a project network
- ⇒ Determines **minimum project completion time**
- ⇒ Delay in critical path delays the entire project

Network Diagram

⇒ Shows sequence and dependency of activities

❖ Types:

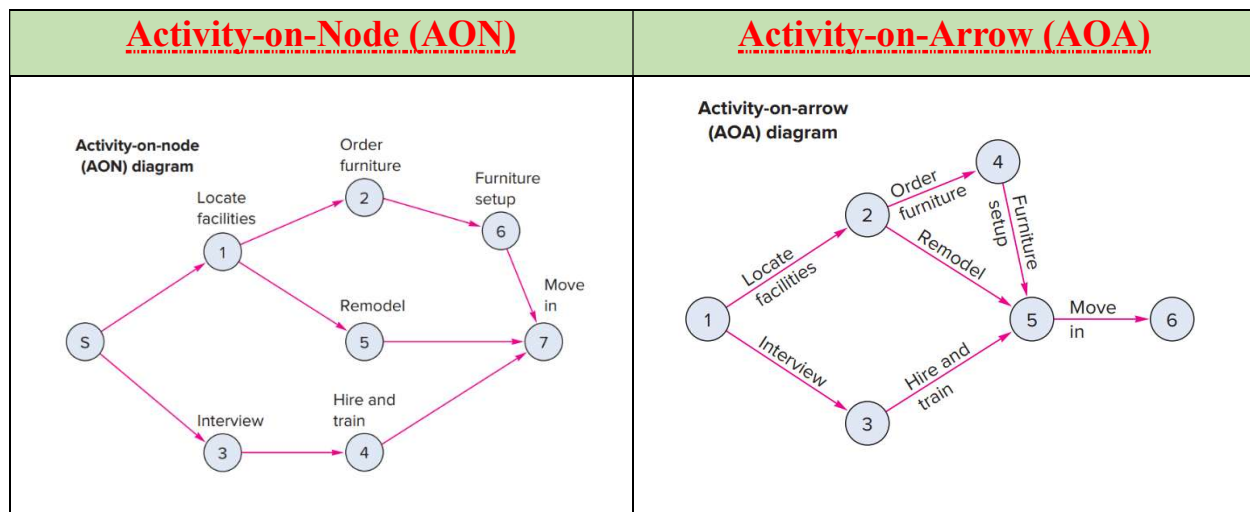
- **Activity-on-Node (AON)** → Activities are nodes
- **Activity-on-Arrow (AOA)** → Activities are arrows



Important Terms

- **Activity:** A task that consumes time and resources
- **Event:** Completion point, consumes no resources
- **Network:** Logical arrangement of activities
- **Path:** Sequence of connected activities
- **Critical Activity:** Activity with zero slack

❖ A Simple Project Network Diagram:

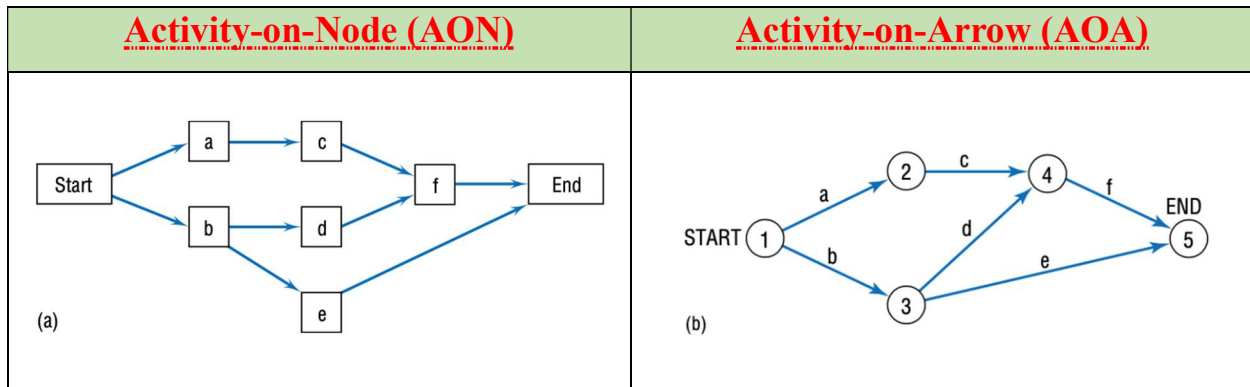


✂ **Problem1:** A project consists of the following activities along with their precedence relationships and durations:

Activity	Immediate Predecessor(s)
A	—
B	—
C	A
D	B
E	B
F	C, D

- Draw the **Activity-on-Node (AON)** network diagram for the project.
- Draw the **Activity-on-Arrow (AOA)** network diagram for the project.
- Identify the **start and end activities** clearly in both diagrams.

❖ **Solutions:**



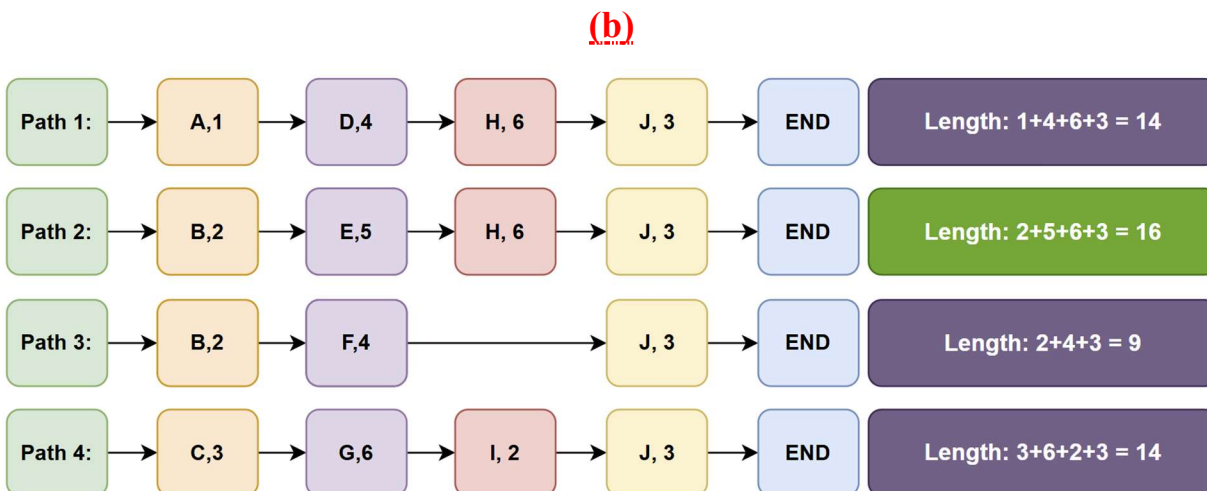
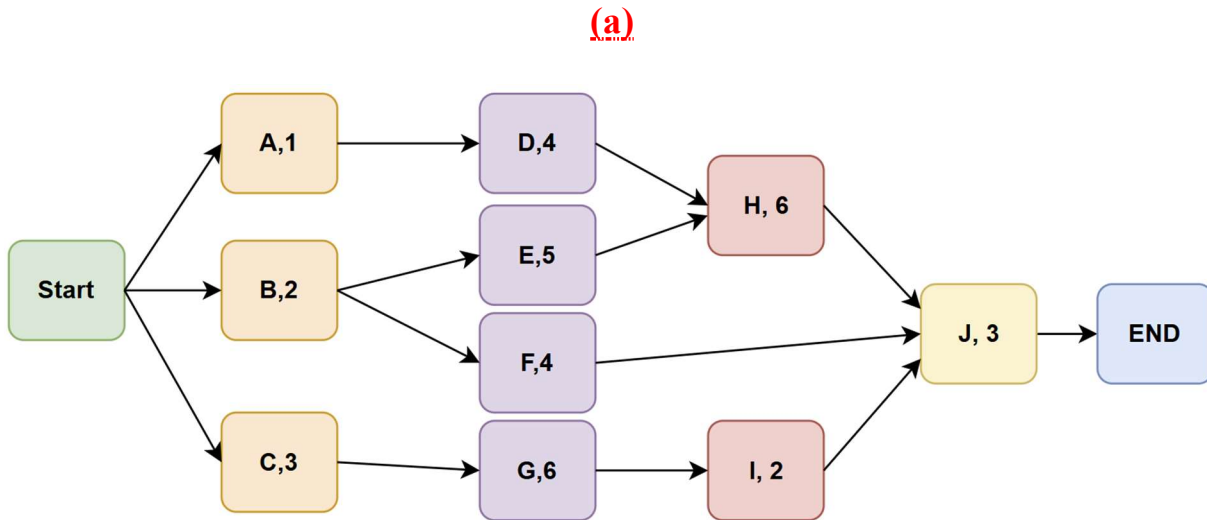
✂ **Problem2:** A project consists of the following activities with their durations and immediate predecessors:

Activity	Duration	Immediate Predecessor(s)
A	1 day	—
B	2 days	—
C	3 days	—
D	4 days	A
E	5 days	B
F	4 days	B
G	6 days	C
H	6 days	D, E
I	2 days	G
J	3 days	H, F, I

a) Draw the **Activity-on-Node (AON)** network diagram for the project.

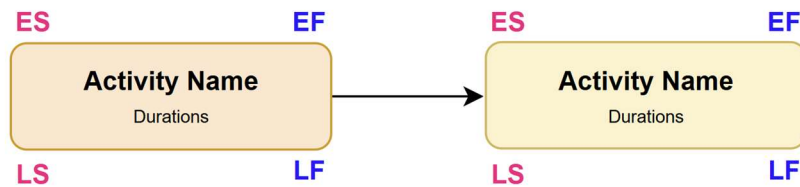
b) Perform **Critical Path Analysis** and determine:

- **Critical Path** of the project



Since, the longest path in a project network determines the critical path of the project, **Path 2** is identified as the critical path of the given project.

Important Terms of Critical Path Analysis



❖ Notes:

ES: Earliest Start

LS: Latest Start

EF: Earliest Finish

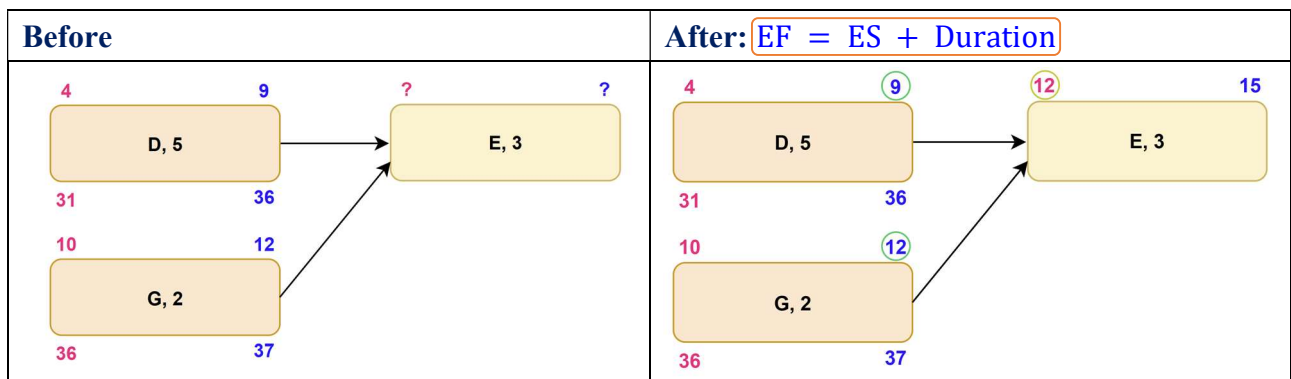
LF: Latest Finish

Forward and Backward Pass

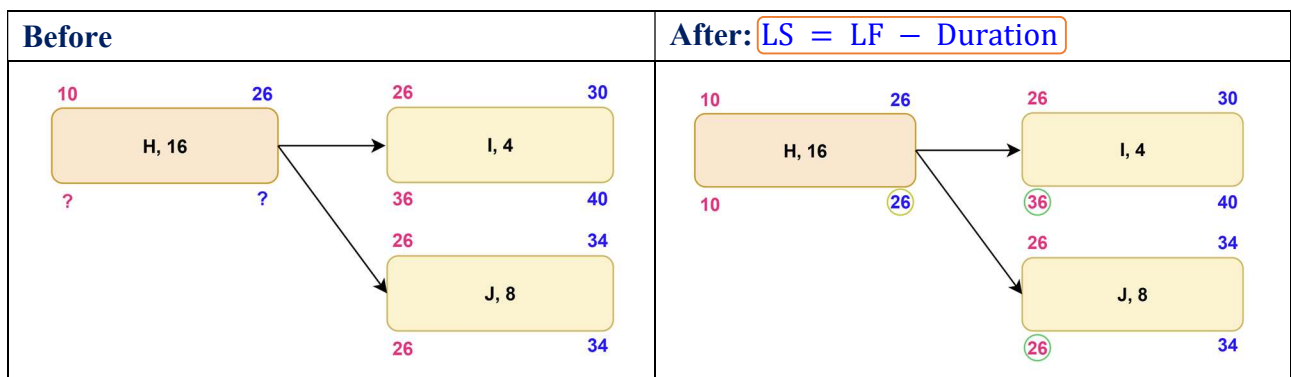
❖ **Forward Pass** → Have to take the **Higher EF** for the Immediate Successor (E)

❖ **Backward Pass** → Have to take the **Lowest LS** for the Immediate Predecessor (H)

➤ **Example: For Forward Pass:**



➤ **Example: For Backward Pass:**



Slack (Float)

⇒ Time an activity can be delayed **without delaying project**

• Formula:

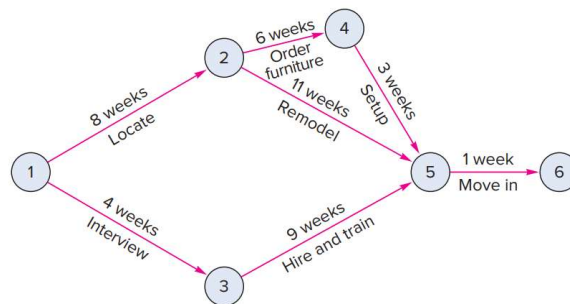
○ $Slack = LS - ES$

○ $Slack = LF - EF$

○ $Path\ Slack = Critical\ Path\ Duration - Path\ Durations$

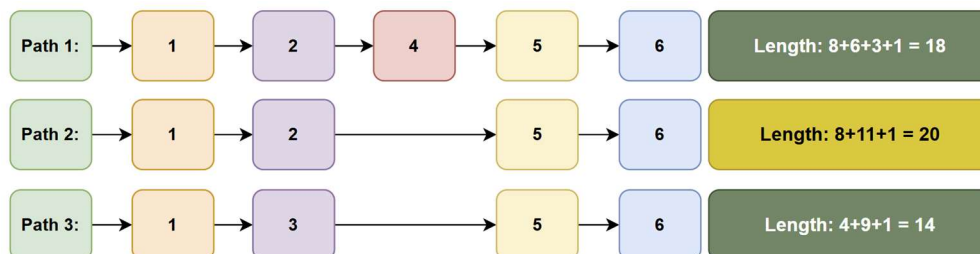
❖ **Notes:** Critical path activities have **zero slack**

🔗 **Problem3:** Given the additional information on the **bank project network** shown in **Figure**, **answer** the following questions assuming **deterministic activity times**:



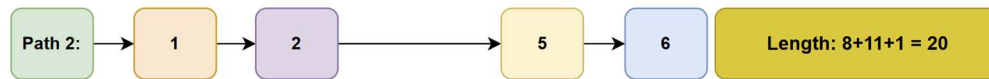
- Determine the **length of each possible path** in the project network.
- Identify the **critical path** of the project.
- Calculate the **expected project completion time**.
- Compute the **slack (float) time for each path**.

(a)



(b)

Since, the longest path in a project network determines the critical path of the project, **Path 2** is identified as the critical path of the given project.



(c)

The **expected project completion time** is equal to the **length of critical path**

Project Completion time = 20 Weeks

(d)

⇒ **Is given,**

path 1 length = 18 weeks

path 2 length = 20 weeks

path 3 length = 14weeks

⇒ **We Know**

Path Slack = Critical Path Duration – Path Durations

⇒ **Path 1 Slack:**

$\text{Path1 Slack} = 20 - 18 = 2 \text{ weeks}$

⇒ **Path 2 Slack:**

$\text{Path2 Slack} = 20 - 20 = 0 \text{ weeks}$

⇒ **Path 3 Slack:**

$\text{Path3 Slack} = 20 - 14 = 6 \text{ weeks}$

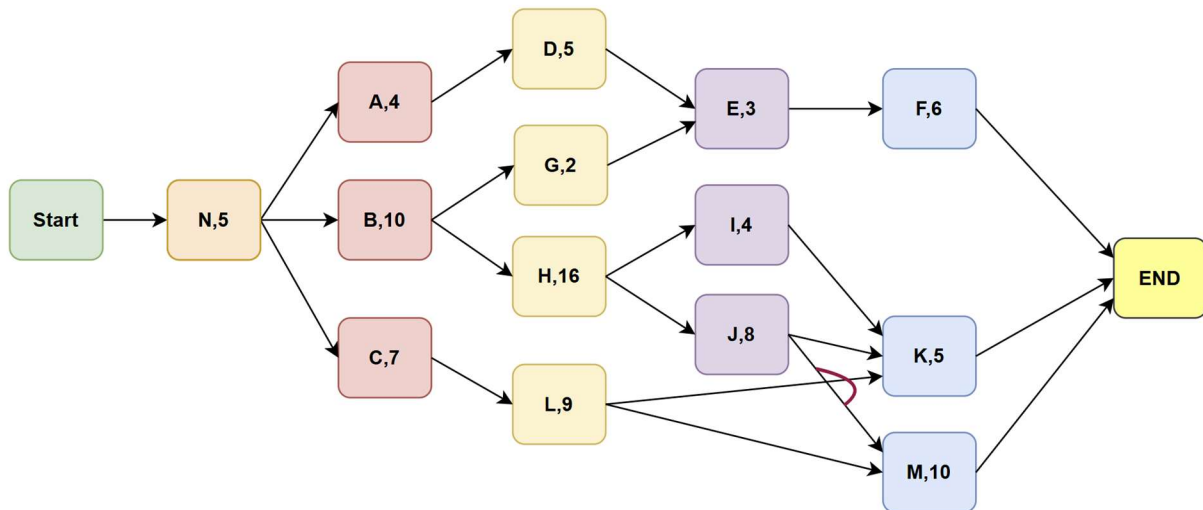
✂ **Problem4:** A project consists of the following activities with their **immediate predecessors** and **durations (in weeks)**:

Activity	Immediate Predecessor(s)	Duration (Weeks)
N	—	5
A	N	4
B	N	10
C	N	7
D	A	5
E	D, G	3
F	E	6
G	B	2
H	B	16
I	H	4
J	H	8
K	I, J, L	5
L	C	9
M	J, L	10

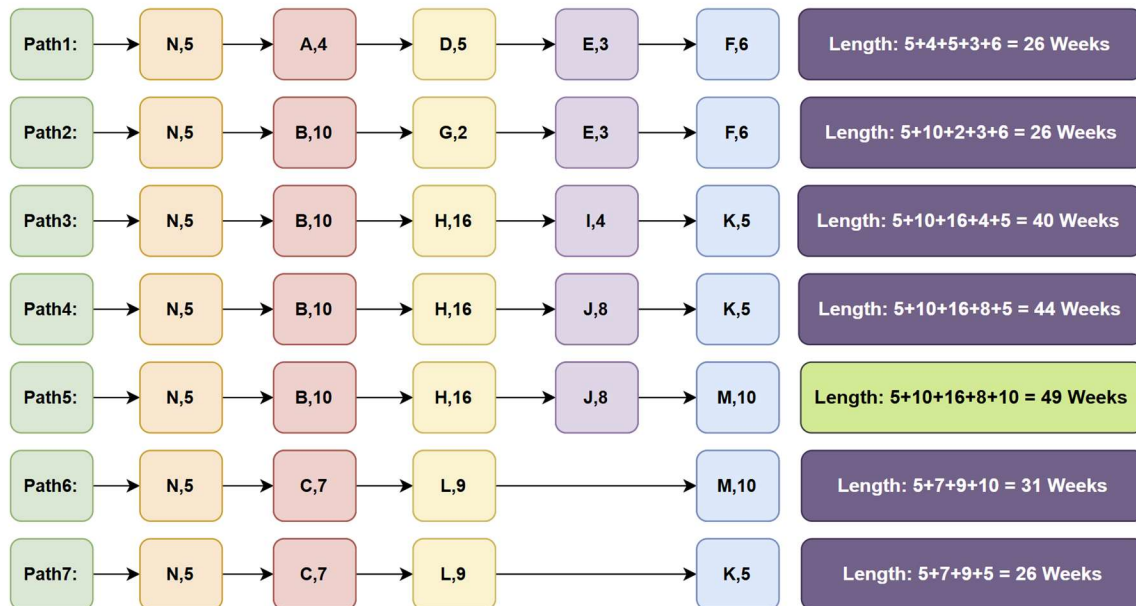
- Draw the **project network diagram** and determine the **critical path** and **project completion time**.
- Calculate the **float (slack)** for **all activities**.
- Can activity **I** be delayed without affecting the project completion time? If yes: i) How many weeks can it be delayed? ii) How much money can be saved if **\$2,500** is saved for each delayed week of activity **I**, while keeping the project completion time unchanged?
- What will be the **impact on the project completion time** if activity **F** is delayed by **2 weeks** and requires an **additional 5 weeks** of duration?
- The Project Manager wants to increase the duration of activity **M** from **10 weeks to 16 weeks**. What will be the **impact on the overall project completion time**?

(a)

❖ Project Network Diagram:



❖ Critical Path:



Since, the longest path in a project network determines the critical path of the project, **Path 5** is identified as the critical path of the given project.

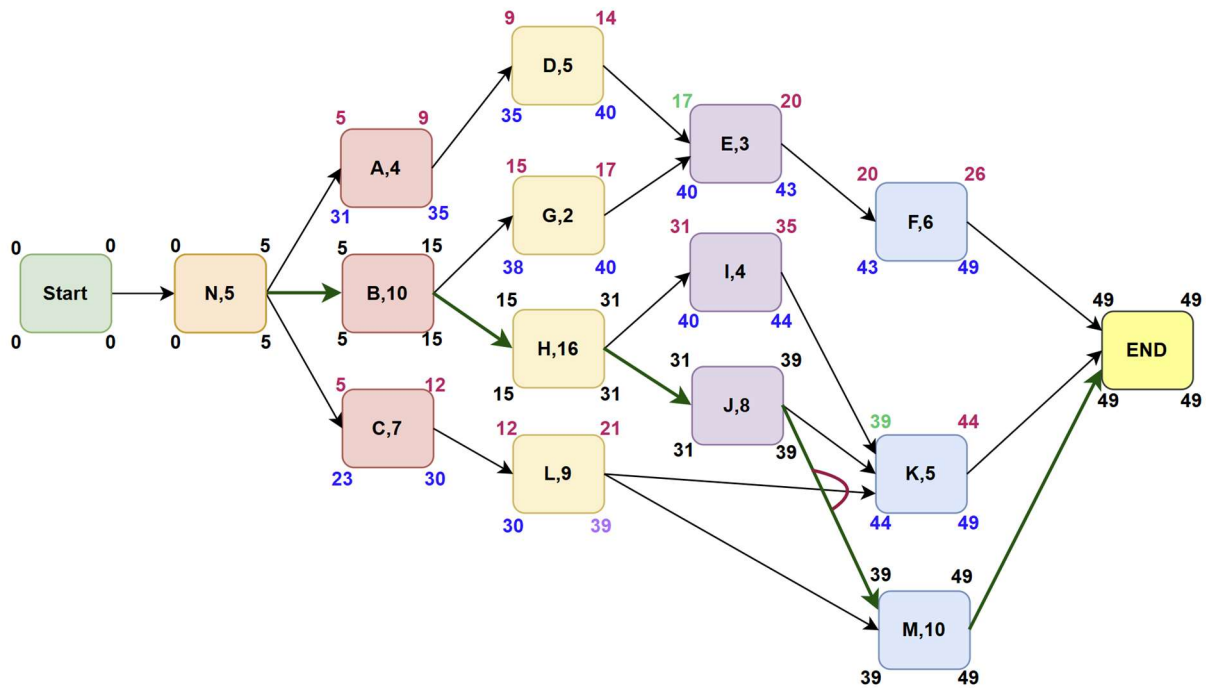
❖ Project Completion Time:

⇒ The **expected project completion time** is equal to the **length of critical path**

Length: $5+10+16+8+10 = 49$ Weeks

(b)

❖ For float (slack):



⇒ We Know:

$$\text{Slack} = LS - ES \parallel LF - EF$$

Slack _A = 31 - 5 = 26	Slack _H = 15 - 15 = 0
Slack _B = 5 - 5 = 0	Slack _I = 40 - 31 = 9
Slack _C = 23 - 5 = 18	Slack _J = 31 - 31 = 0
Slack _D = 35 - 9 = 26	Slack _K = 44 - 39 = 5
Slack _E = 40 - 17 = 23	Slack _L = 30 - 12 = 18
Slack _F = 43 - 20 = 23	Slack _M = 39 - 39 = 0
Slack _G = 38 - 15 = 23	Slack _N = 0 - 0 = 0

(c)

Since, it is a non-critical activity. **So, Yes**, the Activity “I” can be delayed. It can be delayed up to the value of slack that is for 9 weeks.

Cost Saving:

$$\text{savaing} = 9 \times 2500 = \$22500$$

(d)

$$\text{Total Delay} = 2 + 5 = 7 \text{ Weeks}$$

Since, activity F is a non-critical activity with a slack value of 23 weeks, it can be delayed without affecting the overall project completion time.

(e)

Since, activity “M” is a critical activity with a slack value of 0 weeks.

So, it cannot be delayed **without affecting the project completion** time. If the Project Manager increases the duration of activity M from 10 weeks to 16 weeks, the entire project will be delayed.

$$\text{Total Project Delay} = 49 + 6 = 55 \text{ Weeks}$$

As a result, the total project duration will increase by 6 weeks, and the new project completion time will be 55 weeks.

✂ **Problem5:** A project consists of the following activities with their **immediate predecessors** and **durations (in weeks)**:

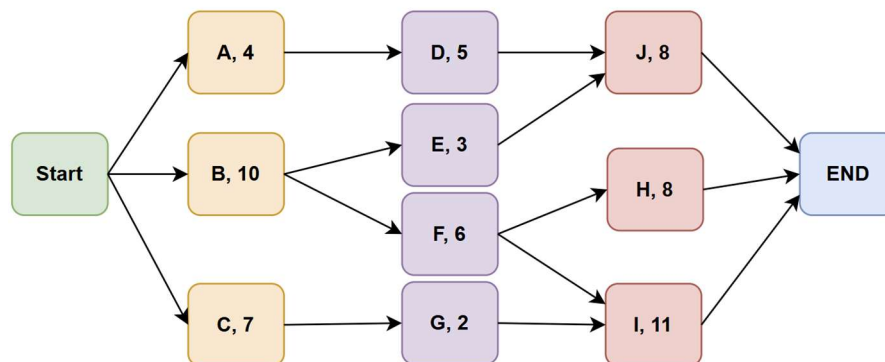
Activity	Immediate Predecessor(s)	Duration (Weeks)
A	—	4
B	—	10
C	—	7
D	A	5
E	B	3
F	B	6
G	C	2
H	F	8
I	F, G	11
J	D, E	8

- Draw the **project network diagram** and determine the **critical path** and **project completion time**.
- Calculate the **float (slack)** for **all activities**.
- If **\$1,000** can be saved for each **delayed week** of activity **H**, how much money can be saved **without changing the project completion time**?

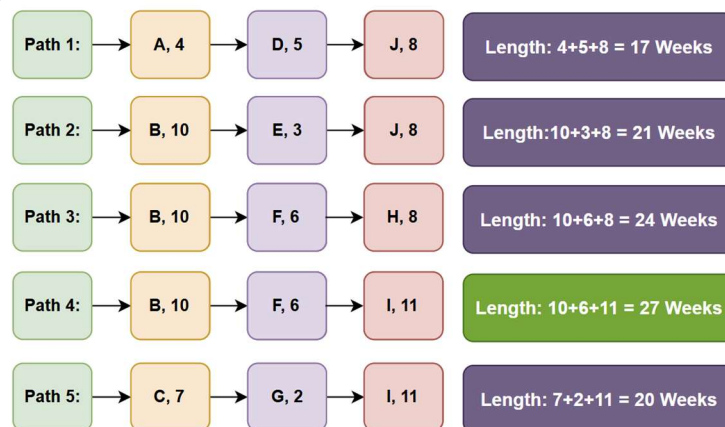
- d) Activity **G** requires a delay of **2 weeks** and an **additional duration of 5 weeks**. Is it possible to accommodate this change **without affecting the project completion time**? Justify your answer.
- e) The Project Manager wants to **shorten the duration of activity I** from **11 weeks to 9 weeks**. What will be the **impact of this change on the project completion time**?

(a)

❖ Project Network Diagram:



❖ Critical Path:



Since, the longest path in a project network determines the critical path of the project, **Path 4** is identified as the critical path of the given project.

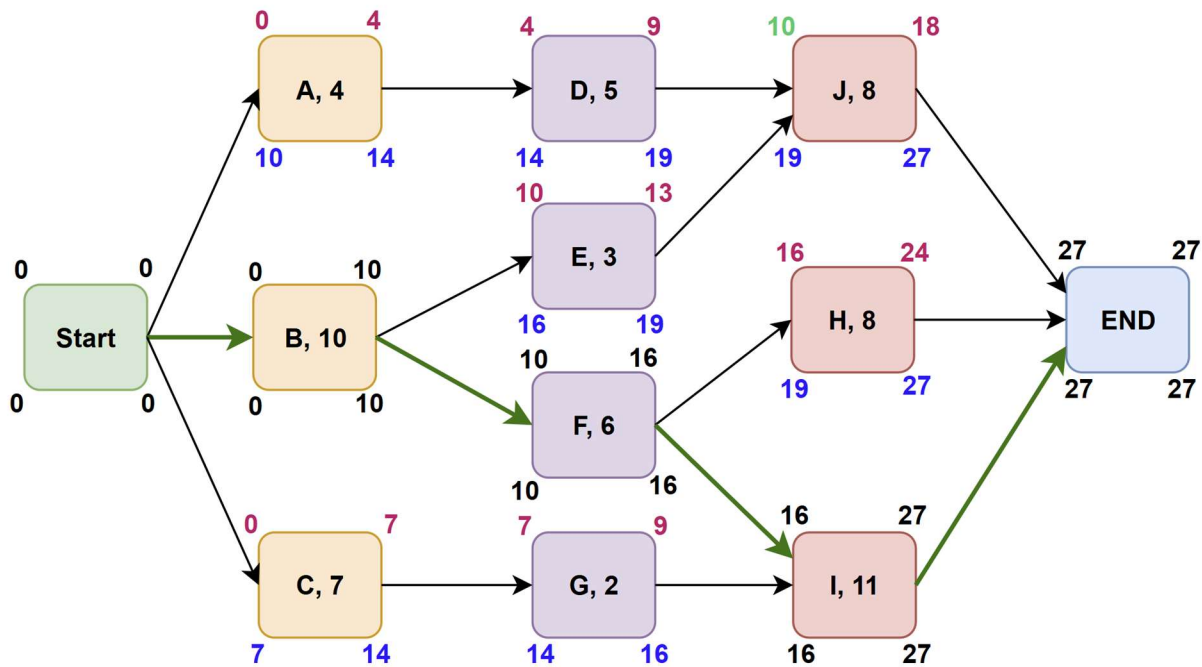
❖ Project Completion Time:

⇒ The **expected project completion time** is equal to the **length of critical path**

Project Completion time: $10+6+11 = 27$ Weeks

(b)

❖ For float (slack):



⇒ **We Know:**

$$\text{Slack} = LS - ES \parallel LF - EF$$

$\text{Slack}_A = 14 - 4 = 10$	$\text{Slack}_F = 10 - 10 = 0$
$\text{Slack}_B = 10 - 10 = 0$	$\text{Slack}_G = 14 - 7 = 7$
$\text{Slack}_C = 14 - 7 = 7$	$\text{Slack}_H = 19 - 16 = 3$
$\text{Slack}_D = 19 - 9 = 10$	$\text{Slack}_I = 16 - 16 = 0$
$\text{Slack}_E = 19 - 13 = 6$	$\text{Slack}_J = 19 - 10 = 9$

(c)

Since, it is a non-critical activity. **So, Yes**, the Activity “H” can be delayed. It can be delayed up to the value of slack that is for 3 weeks.

Cost Saving:

$$\text{savaing} = 3 \times 1000 = \$3000$$

(d)

$$\text{Total Delay} = 2 + 5 = 7 \text{ Weeks}$$

Since, activity **G** is a non-critical activity with a slack value of 7 weeks, it can be delayed without affecting the overall project completion time.

(e)

Since, activity “**I**” is a critical activity with a slack value of 0 weeks.

So, any change in its duration will directly affect the project completion time. If the Project Manager reduces the duration of activity **I** from 11 weeks to 9 weeks, the total project duration will decrease by 2 weeks.

$$\text{Total Project Delay} = 27 - 2 = 25 \text{ Weeks}$$

As a result, the total project duration will decrease by 2 weeks, and the new project completion time will be 25 weeks.

✂ **Problem6:** A project consists of the following activities with their durations and immediate predecessors:

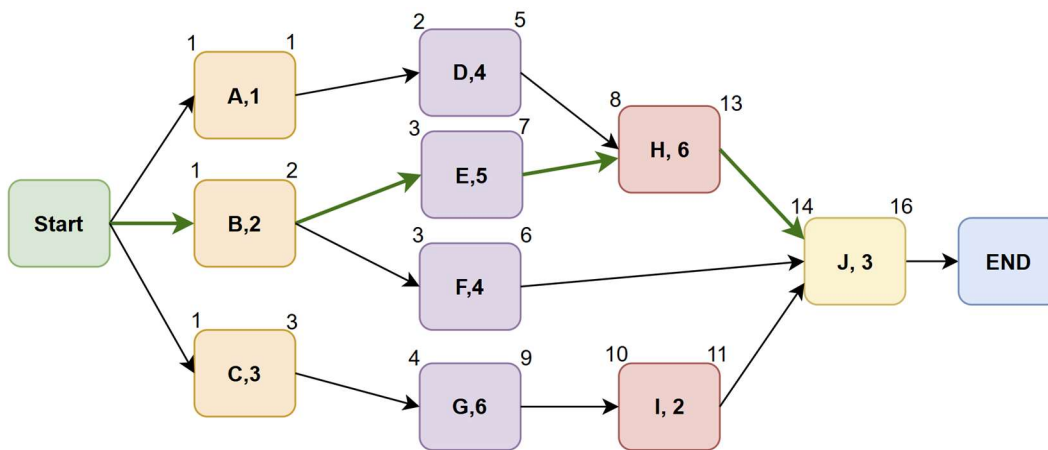
Activity	Duration	Immediate Predecessor(s)
A	1 day	—
B	2 days	—
C	3 days	—
D	4 days	A
E	5 days	B
F	4 days	B
G	6 days	C
H	6 days	D, E
I	2 days	G
J	3 days	H, F, I

a) Draw the **Activity-on-Node (AON)** network diagram for the project.

b) Perform **Critical Path Analysis** and determine:

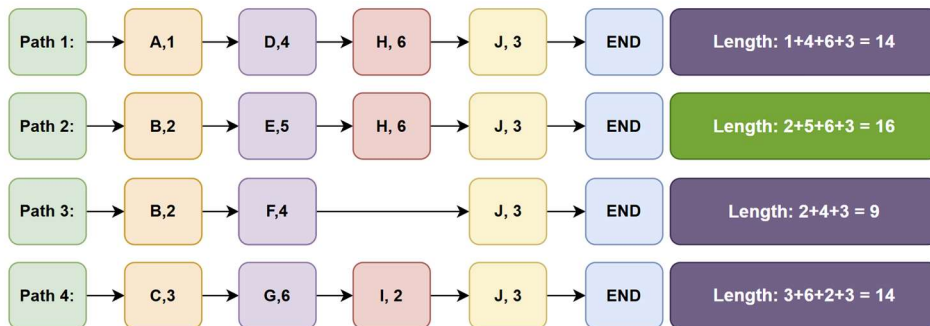
- **Critical Path** of the project
- **Draw** the Gantt Chart of the project

(a)



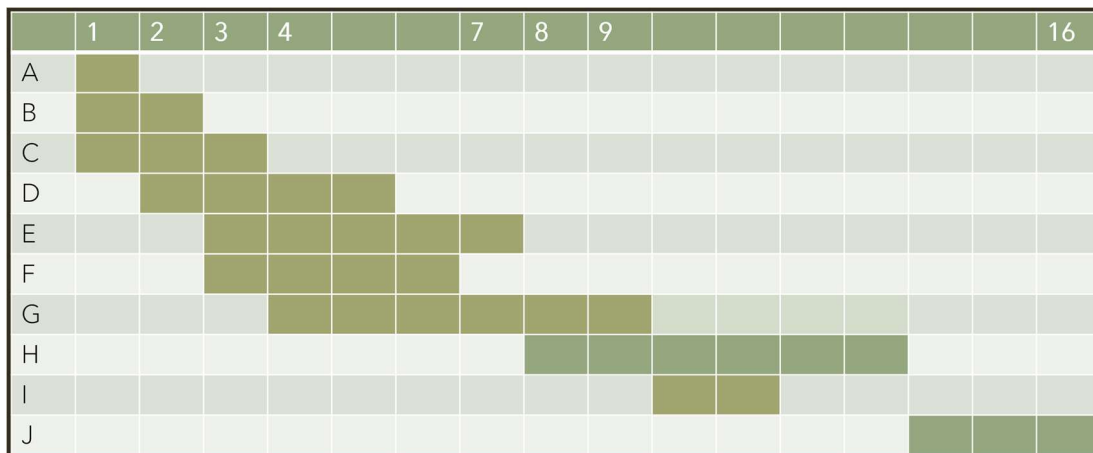
(b)

❖ Now Critical path:



Since, the longest path in a project network determines the critical path of the project, **Path 2** is identified as the critical path of the given project.

❖ Now Gantt chart:



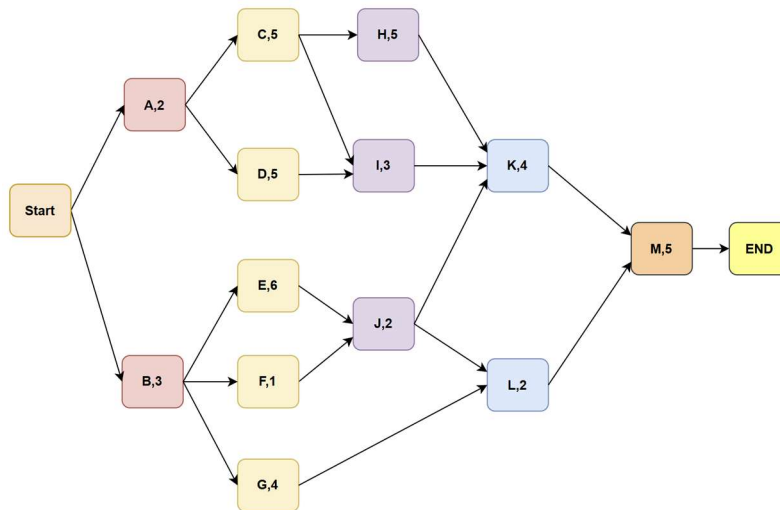
✂ **Problem7:** You have recently appointed as a project manager to make a feasibility study to test **a tidal type electric power generation in Saint-Martin Island**. You have prepared list of activities and estimated duration time, which shows as follows:

Activity code	Duration (week)	Predecessor Activity (code)
A	2	-
B	3	-
C	5	A
D	5	A
E	6	B
F	1	B
G	4	B
H	5	C
I	3	C,D
J	2	E,F
K	4	H,I,J
L	2	J,G
M	5	K,L

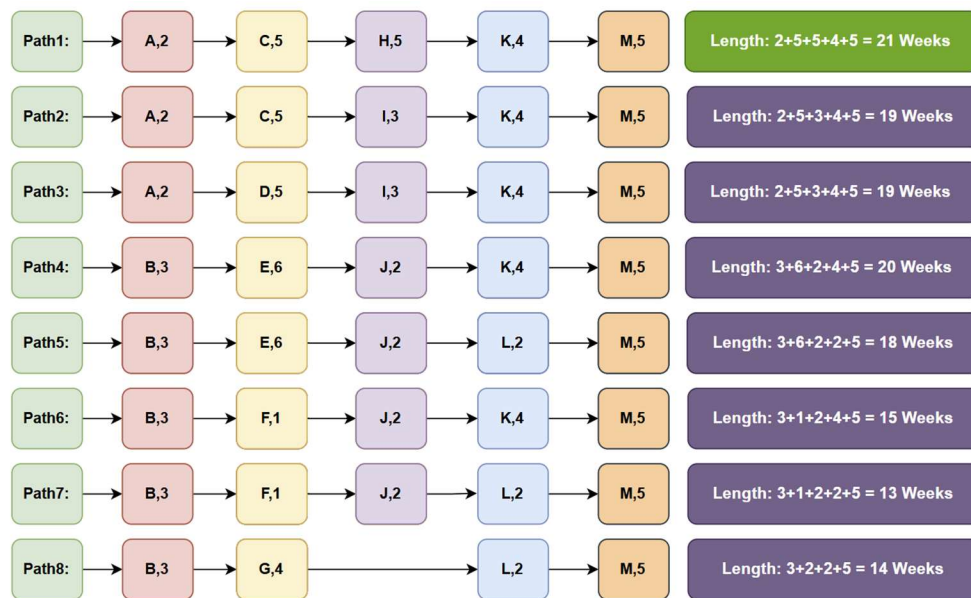
- Calculate how many minimum weeks it will require to complete the project
- What will be the slack time of each activities?
- Due to resource constraint, your manager notified you that activities D, F and G can will only have 50% of resources (i.e. duration time will be doubled). Will it make the project any delay?
- By completing the project on due time, the company will receive an award from project sponsors' of 5,00,000 BDT. If you feel that project will be delayed due to 50% pace of activities D, F and G you have alternate option to hire additional workers from a third party. By hiring additional resources for activities D, F and G to complete in due time it will cost **50,000 BDT each week for each activities**. What will be the minimum additional cost you could allocate to hire resources from third party?

(a)

❖ Project Network Diagram:



❖ Critical Path:



Since, the longest path in a project network determines the critical path of the project, **Path 1** is identified as the critical path of the given project.

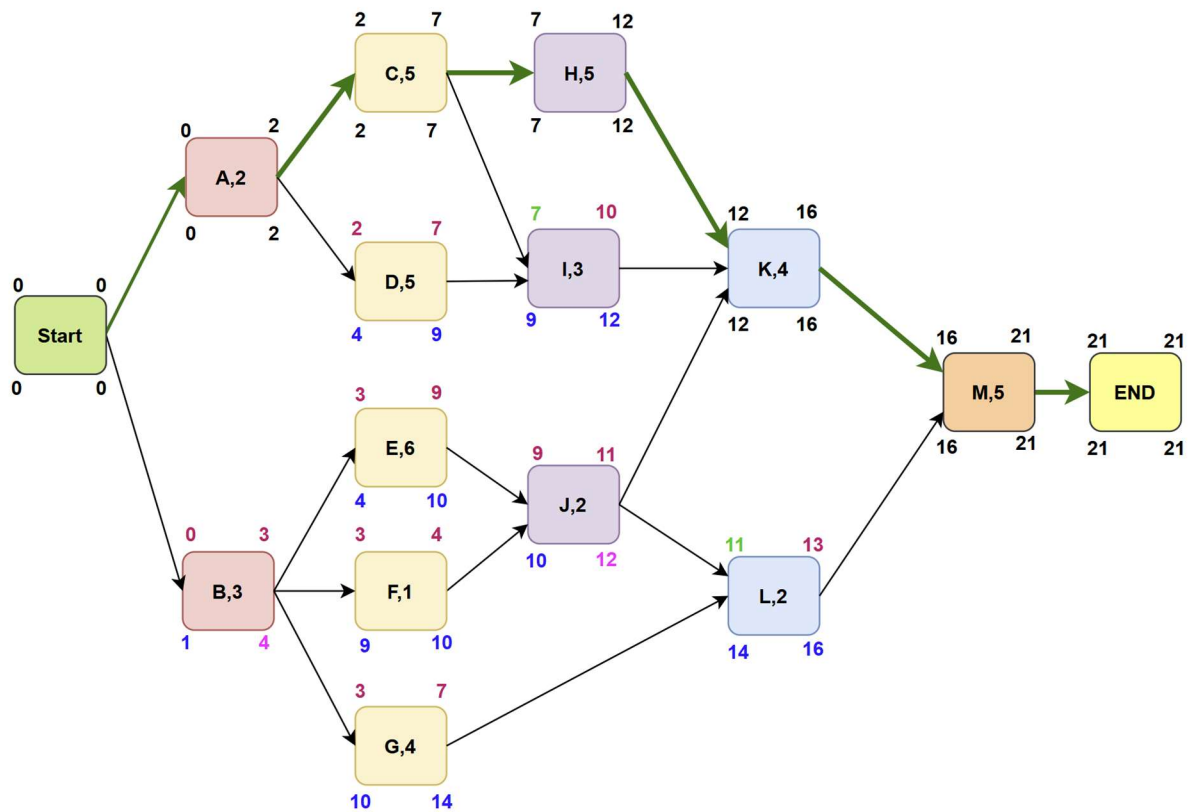
❖ Project Completion Time:

⇒ The **expected project completion time** is equal to the **length of critical path**

Length: $2+5+5+4+5 = 21$ Weeks

(b)

❖ For float (slack):



⇒ We Know:

$$\text{Slack} = LS - ES \parallel LF - EF$$

Slack _A = 2 - 2 = 0	Slack _G = 14 - 7 = 7
Slack _B = 4 - 3 = 1	Slack _H = 12 - 12 = 0
Slack _C = 7 - 7 = 0	Slack _I = 12 - 10 = 2
Slack _D = 4 - 2 = 2	Slack _J = 12 - 11 = 1
Slack _E = 10 - 9 = 1	Slack _K = 16 - 16 = 0
Slack _F = 10 - 4 = 6	Slack _L = 16 - 3 = 3
Slack _M = 16 - 16 = 0	

(c)

➤ **Due to 50% resource availability, the durations of the following activities are doubled:**

- **D: 5 → 10 weeks** (increase = +5 weeks)
- **F: 1 → 2 weeks** (increase = +1 week)
- **G: 4 → 8 weeks** (increase = +4 weeks)

❖ **Comparison with Slack Time:**

Activity	Slack (weeks)	Increase in Duration	Effect
D	2	+5	Exceeds slack
F	6	+1	Within slack
G	7	+4	Within slack

Since, Activity “D, F & G” are non-critical activity so, this activity can be delayed up to their slack value. Without effecting the project completion time.

- Activity **F** can be delayed without affecting the project duration because its increased duration is **within its slack**.
- Activity **G** also remains **non-critical**, since its increased duration **does not exceed its slack**.
- Activity **D**, however, **exceeds its available slack** and therefore **becomes critical**.

❖ **Revised Critical Path**

Because of the delay in **D**, the new critical path becomes: **A → D → I → K → M**

Project duration: **2 + 10 + 3 + 4 + 5 = 24 weeks**

So, the project will be delayed = **24 – 21 = 3 weeks**.

⇒ **Yes, the project will be delayed by 3 weeks.** This delay occurs because the increased duration of **activity D exceeds its slack time**, causing it to become part of the **new critical path**, while activities **F and G do not affect the project duration**.

(d)

➤ **Minimum Additional Cost to Complete the Project on Due Time:**

- Original project duration (**due time**) = **21 weeks**
- New project duration after resource reduction = **24 weeks**
- Total delay = **3 weeks**

➤ **From part (c), only activity D causes the delay because:**

- Its **duration increase (5 weeks)** exceeds its **slack (2 weeks)**
- Activities **F and G remain non-critical**

so, to complete the project **within the original due time**, we only need to **crash activity D**.

❖ **Cost of Hiring Additional Resources**

- Cost = **50,000 BDT per week per activity**
- Weeks to be reduced = **3 weeks**
- Activity to be crashed = **D only**

$$\text{Minimum additional cost} = 3 \times 50,000 = 150,000 \text{ BDT}$$

⇒ By investing **1,50,000 BDT**, the company can complete the project within the original **21-week deadline** and still receive the sponsor's award of **5,00,000 BDT**.

So, net additional profit earned will be:

$$5,00,000 - 1,50,000 = 3,50,000 \text{ BDT}$$

⇒ By allocating a minimum additional cost of **1,50,000 BDT**, the company avoids project delay and gains an extra profit of **3,50,000 BDT**.

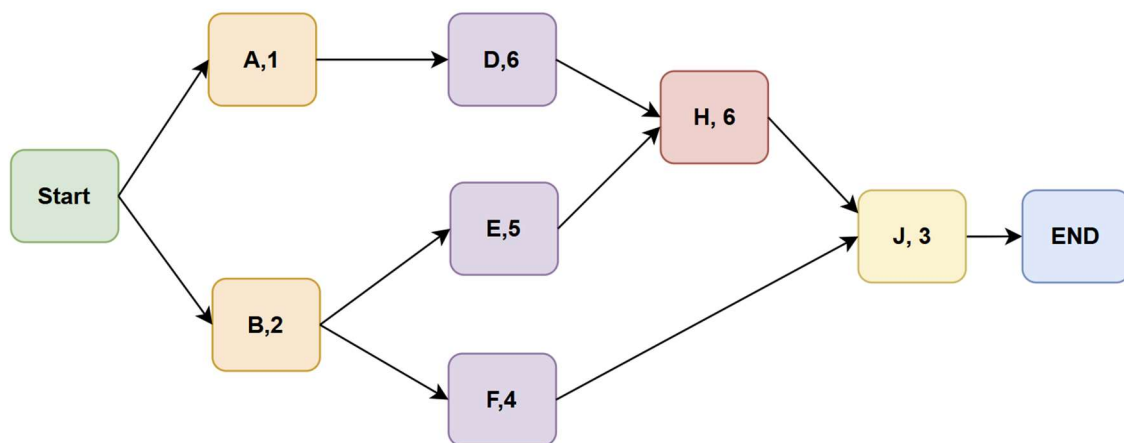
✂ **Problem8:** A project consists of the following activities with their durations and immediate predecessors:

Activity	Duration	Immediate Predecessor(s)
A	1 day	—
B	2 days	—
D	6 days	A
E	5 days	B
F	4 days	B
H	6 days	D, E
J	3 days	H, F, I

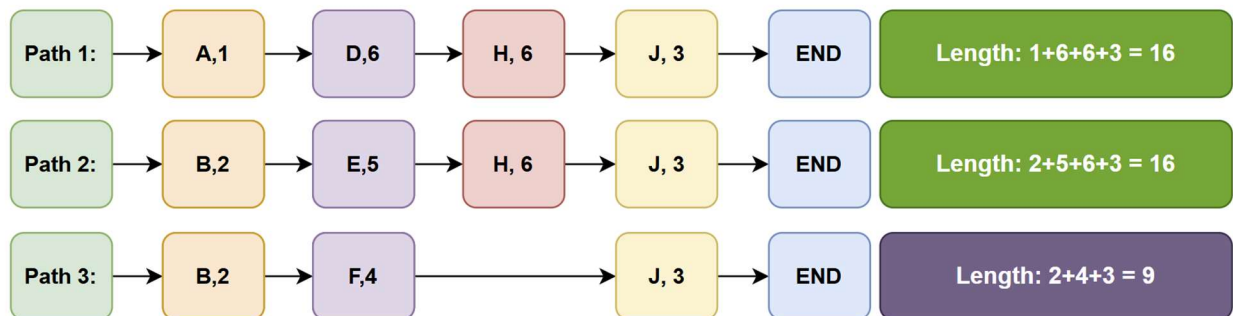
- Draw the **project network diagram** and determine the **critical path** and **project completion time**.
- Calculate the **float (slack)** for all activities.
- The Project Manager wants to **Increase the duration of activity H for 2 days & J for 1 day**. What will be the **impact of this change on the project completion time**?

(a)

❖ Project Network Diagram:



❖ Now Critical path:



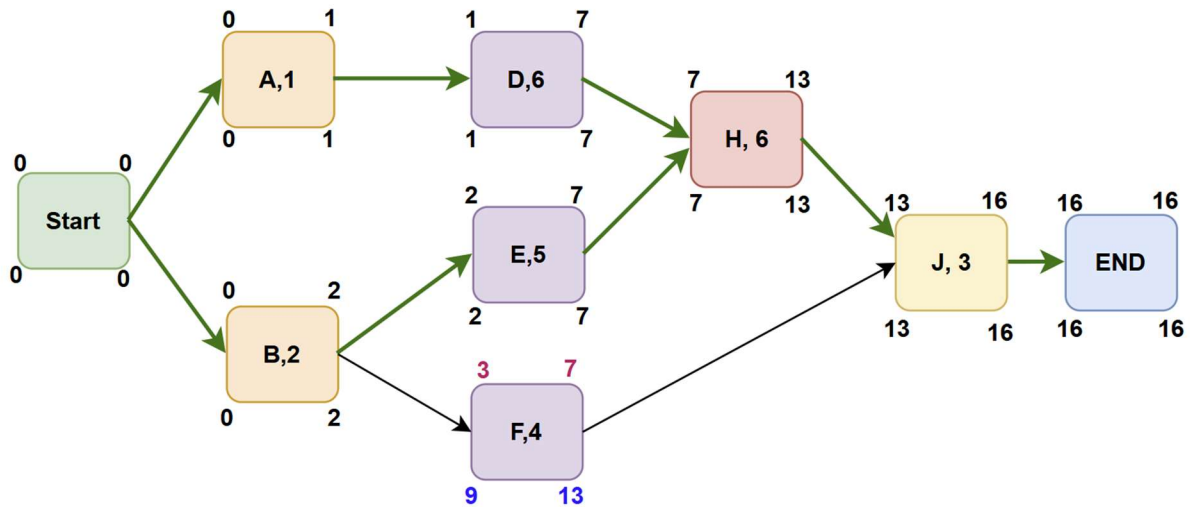
Since, the longest path in a project network determines the critical path of the project, **Path 1 & Path 2** is identified as the critical path of the given project.

❖ Project Completion Time:

⇒ The **expected project completion time** is equal to the **length of critical path**

Project Completion Time = 16 days

(b)



⇒ **We Know:**

$$\text{Slack} = LS - ES \parallel LF - EF$$

Slack _A = 0 - 0 = 0	Slack _E = 2 - 2 = 0
Slack _B = 0 - 0 = 0	Slack _F = 9 - 3 = 6
Slack _D = 1 - 1 = 0	Slack _H = 13 - 13 = 0
Slack _I = 16 - 16 = 0	

(c)

Since, activity “H” & “J” is a critical activity with a slack value of 0 weeks.

So, any change in its duration will directly affect the project completion time. If the Project Manager increased the duration of activity H for 2 days & J for 1 day.

$$\text{Total Project Delay} = 16 + 2 + 1 = 19 \text{ days}$$

As a result, the total project duration will increased by 3 days, and the new project completion time will be 19 days.

TOTAL QUALITY MANAGEMENT

✂ **Problem1:** Construct a **Pareto diagram** from the data given in the table below showing different types of medication errors.

Category	Frequency
Wrong medicine	15
Medicine dc'd	4
Wrong dose	100
Wrong patient	8
Wrong time	70
Missed dose	3
Grand Total	200

➤ **Solutions:**

Step 1: Arrange Categories in **Descending Order**.
(Pareto rule: highest frequency first)

Category	Frequency
Wrong dose	100
Wrong time	70
Wrong medicine	15
Wrong patient	8
Medicine dc'd	4
Missed dose	3

Step 2: Calculate **Percentage** for Each Category

$$\text{Percentage} = \frac{\text{Frequency}}{\text{Total}} \times 100$$

Category	Frequency	Percentage
Wrong dose	100	50%
Wrong time	70	35%
Wrong medicine	15	7.5%
Wrong patient	8	4%
Medicine dc'd	4	2%
Missed dose	3	1.5%

Step 3: Calculate **Cumulative Percentage** (Add percentages one by one)

Category	Frequency	Percentage	Cumulative %
Wrong dose	100	50%	50%
Wrong time	70	35%	85%
Wrong medicine	15	7.5%	92.5%
Wrong patient	8	4%	96%
Medicine dc'd	4	2%	98.5%
Missed dose	3	1.5%	100%

Step 4: Pareto Diagram

